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Knowledge Discovery From Free Text

 Final-Thesis
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Document Details

Submission ID**trn:oid:::1:3571551491****Submission Date****May 16, 2026, 7:31 PM GMT+3****Download Date****May 19, 2026, 11:26 AM GMT+3****File Name****thesis.docx****File Size****119.9 KB****78 Pages****24,398 Words****144,932 Characters**

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Addis Ababa University

College of Natural Sciences

Knowledge Discovery from Free Text: A BERT-Based System for Extracting Violent Event Information from African News Reports

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A Thesis Submitted to the Department of Computer Science in Partial Fulfillment for the Degree of Master of Science in Computer Science

Addis Ababa, Ethiopia

May 2026

Addis Ababa University

College of Natural Sciences

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This is to certify that the thesis prepared by Binalfew Kassa Mekonnen, titled: *Knowledge Discovery from Free Text: A BERT-Based System for Extracting Violent Event Information from African News Reports* and submitted in partial fulfillment of the requirements for the Degree of Master of Science in Computer Science complies with the regulations of the University and meets the accepted standards with respect to originality and quality.

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Abstract

The volume of unstructured news reporting on violent events in Africa exceeds analyst capacity to read, code, and act upon within operationally useful timeframes, and at the African Union Continental Early Warning System this transformation remains predominantly manual. This thesis presents VioNER, an end-to-end system that extracts 5W1H attributes (Who, What, Where, When, Whom, How) from African news reports of violent events through fine-tuned BERT-based named entity recognition coupled with a knowledge-base validation layer. A grounded annotation schema of eight entity types (ACTOR, VICTIM, ACTION, DATE, REGION, CITY, DISTRICT, CASUALTIES) is introduced in Beginning-Inside-Outside (BIO) format, alongside a four-level hierarchical taxonomy of approximately ninety-five African violent-event categories. The model was fine-tuned on a fifty-thousand-example corpus derived from the Armed Conflict Location and Event Data project, combining stratified diversity sampling with template-based augmentation to address severe class imbalance, and using a focal-loss objective with inverse-frequency class weights to counter the dominance of the O label. The fine-tuned bert-base-cased model achieved a macro F1 of 0.887 and a micro F1 of 0.909 on the held-out validation set, converging in two epochs; an ablation showed that focal loss with class weighting raised the rarest entity (VICTIM) by eleven F1 points over plain cross entropy. The trained model is exposed through a FastAPI service, a PostgreSQL event store, a curated knowledge base of African armed groups and cities, and a React web application for training, inference, event management, and analytics. The artefact substantially reduces analyst processing time and produces structured records suitable for downstream early-warning analysis.

Keywords: Named Entity Recognition, BERT, Event Extraction, Violent Events, African Conflicts, 5W1H, Focal Loss

Dedication

TBD

Acknowledgements

I am deeply indebted to my advisor, Dr. Fekade Getrahun, for his guidance, critical reading, and steady encouragement throughout this research. His insistence on rigour and operational relevance shaped every chapter of this thesis.

I am grateful to the staff of the Addis Ababa University Department of Computer Science for their support during the programme, and to the African Union Continental Early Warning System (AU-CEWS) for articulating the operational requirements that motivated this work. I also thank the Armed Conflict Location and Event Data Project (ACLED) for maintaining the open dataset on which this study relies.

I acknowledge the prior thesis work of Taye Abdulkadir, whose exploration of 5W extraction in the African context provided a foundation that this research extends.

Finally, I thank my family and friends for their patience and support during the long months of training, debugging, and writing.

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Acronyms and Abbreviations

Acronym	Expanded form
5W1H	Who, What, Where, When, Whom, How
ACLED	Armed Conflict Location and Event Data Project
ADF	Allied Democratic Forces
API	Application Programming Interface
AQIM	Al-Qaeda in the Islamic Maghreb
AU	African Union
AU-CEWS	African Union Continental Early Warning System
BERT	Bidirectional Encoder Representations from Transformers
BIO	Beginning, Inside, Outside (tagging scheme)
CAMEO	Conflict and Mediation Event Observations
CAR	Central African Republic
CRF	Conditional Random Field
CUDA	Compute Unified Device Architecture
DRC	Democratic Republic of the Congo
ECOWAS	Economic Community of West African States
F1	Harmonic mean of precision and recall
FARDC	Forces Armées de la République Démocratique du Congo
GDELT	Global Database of Events, Language, and Tone
IED	Improvised Explosive Device
IGAD	Intergovernmental Authority on Development
JNIM	Jama'at Nasr al-Islam wal Muslimin
JWT	JSON Web Token
KB	Knowledge Base
LRA	Lord's Resistance Army
LSTM	Long Short-Term Memory
ML	Machine Learning
MPS	Metal Performance Shaders

Acronym	Expanded form
NER	Named Entity Recognition
NEXUS	News cluster Event eXtraction Utilizing language Structures
NLP	Natural Language Processing
NOEM	News Ontology Event Model
OWL	Web Ontology Language
PMVE	Politically Motivated Violent Events (ontology)
PoS	Part of Speech
RDF	Resource Description Framework
REC	Regional Economic Community
REST	Representational State Transfer
RoBERTa	Robustly Optimised BERT Approach
RSF	Rapid Support Forces
SPARQL	SPARQL Protocol and RDF Query Language
SQL	Structured Query Language
SVBIED	Suicide Vehicle-Borne Improvised Explosive Device
UCDP	Uppsala Conflict Data Program
UI	User Interface
VBIED	Vehicle-Borne Improvised Explosive Device
VioNER	Violent Event Named Entity Recognition (this work)

1. Introduction

The research problem addressed by this thesis sits at the intersection of three pressures: a flood of African violent-event reporting in unstructured news text, an analyst workforce that cannot read it fast enough, and an early-warning operational requirement that depends on the timely conversion of that text into structured records. The sections that follow lay out the background, motivation, problem statement, objectives, and methods that frame the rest of the work, and close with a roadmap to the remaining chapters.

1.1 Background

The volume of digital news content describing violent events on the African continent has grown faster than the human capacity to read, code, and act upon it. Reports of attacks, clashes, raids, bombings, displacement, and other forms of armed and political violence are disseminated through wire services, mainstream outlets, regional newspapers, and increasingly through online media [1]. The overwhelming majority of this content is unstructured natural-language text intended for human consumption rather than machine processing. Converting these narratives into structured, queryable representations is a

prerequisite for systematic monitoring, statistical aggregation, and evidence-based decision making.

Natural Language Processing (NLP) is the half-century-old discipline that attempts to make computers do useful things with human language, and the toolbox it has developed — part-of-speech tagging, syntactic parsing, named entity recognition, coreference resolution [2] — is exactly what an early-warning analyst is doing implicitly when reading a news article. Information Extraction (IE) is the specific sub-discipline that tries to mechanise the analyst's work: take prose in, produce structured records out. Event Extraction, in turn, narrows IE to events — verbs, nominalised actions, and their participants — and asks the model to recover who did what to whom, where, when, and how, from text alone [3], [4].

The 5W1H frame I adopt is not new. Journalism schools have taught it for over a century, and analysts in early-warning centres read for those slots whether or not they explicitly call them that. The attractive property for an automated system is that 5W1H decouples *what to extract* from *which event types exist*: even before any taxonomic question is settled, an extractor can populate a 5W1H record that downstream analytics can later classify.

The technical opportunity is recent. Vaswani and colleagues' transformer architecture [5] and the line of pre-trained language models built on it — most prominently BERT [6] — narrowed the gap between research prototypes and operational NLP systems for token-level tasks by an order of magnitude in five years [7]. The combination of large-scale self-supervised pre-training with small-scale task-specific fine-tuning is what makes a 50,000-example domain corpus a workable input for a competitive NER model, where ten years earlier it would not have been.

Within the specific domain of conflict and violence monitoring, several long-running data collection efforts have constructed structured event databases by human coding. The Armed Conflict Location and Event Data Project (ACLED) is the most influential of these for the African context, providing georeferenced records of political violence and protest events with detailed actor, location, and fatality information [8]. The Uppsala Conflict Data Program (UCDP) [9] and the Global Database of Events, Language, and Tone (GDELT) [10] play complementary roles. While these initiatives have demonstrated the value of structured event data, the underlying collection processes remain heavily manual or rely on coarse-grained automation that may miss nuance or African-specific patterns.

The African Union Continental Early Warning System (AU-CEWS) operates within this landscape as a continental mechanism for monitoring conflict and crisis dynamics across the African Union member states. AU-CEWS aggregates news content through its Africa Media Monitor tool and draws on it for situation analysis, briefings, and early warning products. The transformation of monitored content into structured, analysable form remains, however, the limiting factor on the system's throughput.

This thesis sits at the intersection of these strands. It uses a fine-tuned BERT model as the core of a 5W1H extraction pipeline; it targets African conflict reporting in particular; and it

organises the outputs around a knowledge base of African armed groups, locations, and a hierarchical taxonomy of violent events. The remainder of this chapter motivates the work in greater detail and states the specific objectives that the work pursues.

1.2 Motivation

Consider a typical morning in a continental monitoring centre. A single analyst is responsible for a region in which, over the preceding twenty-four hours, between two hundred and four hundred news items were aggregated by the centre's media-monitoring tool. Of these, perhaps fifteen to forty describe violent incidents that the analyst must read, structure into actor / action / location / time / casualty fields, classify, and route to the appropriate desk. By mid-morning the analyst has reached perhaps a quarter of the queue; by close of business, the residue rolls over into the next day. This pattern repeats across regions and analysts. The binding constraint on continental situation awareness is not the sophistication of analytical models but the throughput of human reading and coding.

My own exposure to this operational context at the AU-CEWS Situation Monitoring Centre, and the internal documentation of its data-collection and analysis tools [38], revealed two persistent gaps. The first is a *throughput gap*: the volume of news that human analysts can read in a working day is a small fraction of the daily inflow, and the cognitive load of reading and coding violent-event reports is high. The second is a *consistency gap*: analysts coding the same article may diverge in how they categorise actors, events, and locations, especially under time pressure or in fast-moving situations. These two gaps interact. A system that produces consistent structured records from a larger fraction of the inflow would free analyst attention for interpretation rather than extraction, and would also raise the comparability of records across analysts, regions, and time periods.

Three further observations motivate the technical choices made in this thesis. First, African conflict reporting features patterns and named entities that are under-represented in generic, off-the-shelf NLP models trained on European or North American news corpora. Generic NER models routinely misclassify African armed groups, regional administrative divisions, and locally significant locations [11]. A domain-specific model fine-tuned on African conflict reporting is therefore likely to outperform off-the-shelf alternatives by a meaningful margin.

Second, the class distribution within a violent-event NER task is extremely skewed. The overwhelming majority of tokens in a typical news sentence carry the outside (O) label, while entity tokens form a small fraction of the total. Naive cross-entropy training on this distribution tends to under-fit minority entity types. Techniques such as focal loss [12] and inverse-frequency class weighting are well-suited to this regime and have been shown to improve recall on rare classes without sacrificing precision on dominant ones.

Third, the operational utility of extracted entities depends not only on per-token accuracy but on whether the extracted records can be trusted, queried, and audited. A knowledge base of known armed groups, conflict-affected cities, weapons, and a hierarchical taxonomy of violent events allows the extracted output to be cross-checked against

curated reference data, with low-confidence or implausible extractions flagged for human review.

The motivation behind this thesis is therefore both academic and practical. Academically, it contributes a fine-tuned, domain-specific BERT model and a corresponding annotated dataset for the African violent-event NER task, together with a hierarchical taxonomy designed for that context. Practically, it provides a deployable web platform that exposes training, inference, event management, and analytics through a unified interface, lowering the technical barrier for operational adoption.

1.3 Statement of the Problem

The central problem addressed in this thesis is the absence, in the African early-warning ecosystem, of an accurate and openly documented pipeline for converting English-language news reports of violent events into structured 5W1H records suitable for downstream analysis. The problem can be decomposed into three sub-problems.

Sub-problem 1: Domain-specific entity recognition. Generic NER models trained on news corpora do not consistently recognise African-specific armed groups, regional administrative units, conflict-affected localities, and the casualty descriptions used in African conflict reporting. Off-the-shelf models also do not distinguish the actor, victim, action, and casualty roles that an operational analyst cares about. A model fine-tuned on annotated African conflict text is therefore required, supported by a grounded annotation schema that captures the operationally relevant entity types and excludes those that cannot be reliably grounded in the source text.

Sub-problem 2: Severe class imbalance. Within the annotated corpus, the distribution of Beginning-Inside-Outside (BIO) labels is heavily dominated by the O label, with entity tokens forming only a minority of the total. Within the entity tokens themselves, ACTOR, CITY, DATE, REGION, and DISTRICT labels dominate, while VICTIM, ACTION, and CASUALTIES are substantially rarer. Naive training tends to over-fit to the dominant classes and under-recover the rare but operationally important ones. A loss function and sampling strategy that compensate for this imbalance are required.

Sub-problem 3: Operational packaging. A trained model alone is not an operational capability. To be useful, the model must be exposed through a documented API, paired with a curated knowledge base, embedded in a workflow that supports article ingestion, event storage, and analytics, and operable by users who are not machine learning specialists. Most prior academic work on event extraction in the African context has stopped at the model boundary, leaving the operational packaging undone. This thesis addresses operational packaging as a first-class research output rather than as implementation detail.

Research Questions

The three sub-problems above are pursued through the following research questions.

1. Which entity types in African violent-event news reports can be reliably grounded in source text, and what is an appropriate BIO schema for fine-tuning a BERT model on those entities?
2. How effectively can a fine-tuned BERT model recognise the chosen entities, and what loss function and sampling strategy produce the most balanced per-entity performance under severe class imbalance?
3. To what extent does a curated knowledge base of African armed groups, conflict locations, and a hierarchical taxonomy improve the trustworthiness and downstream utility of the extracted records?
4. What system architecture allows the model, the knowledge base, and the analytics layer to be operated together by users without machine learning expertise?

Significance of the Study

This study is significant on three grounds.

Operational significance. The artefact reduces the analyst time required to convert raw news into structured event records. In the target context (AU-CEWS Situation Monitoring), analyst time is the binding constraint on situation awareness during fast-moving crises. Even a partial reduction in the cost of structured event extraction translates directly into faster, broader, and more consistent monitoring.

Methodological significance. The combination of a grounded entity schema, focal-loss training under severe class imbalance, and a curated domain knowledge base is, to the best of my knowledge, the first such combination applied to African violent-event extraction at this scale. The methodology is documented in sufficient detail to be reproducible by other researchers.

Resource significance. The four-level taxonomy of African violent events, the entity annotation schema, and the curated knowledge base of African armed groups and conflict-affected cities are themselves reusable artefacts. They can be adopted or adapted by researchers and practitioners working on adjacent tasks in conflict monitoring, humanitarian protection, and security analysis.

1.4 Objectives

General Objective

The general objective of this research is to design, implement, and evaluate VioNER, an integrated system that extracts 5W1H attributes of violent events from African English-language news reports through fine-tuned BERT-based named entity recognition, supports those extractions with a curated knowledge base and hierarchical taxonomy, and exposes the full capability through a documented web platform.

Specific Objectives

The following specific objectives operationalise the general objective.

1. Review the literature on information extraction, event extraction, named entity recognition, transformer-based language models, and conflict event coding, and position the work with respect to it.
2. Define a grounded eight-entity BIO schema (ACTOR, VICTIM, ACTION, DATE, REGION, CITY, DISTRICT, CASUALTIES) and document inclusion and exclusion rules for each.
3. Construct a four-level hierarchical taxonomy of violent events tailored to African conflict patterns, with approximately ninety five terminal categories and explicit decision rules for ambiguous cases.
4. Assemble a training corpus of approximately fifty thousand examples derived from ACLED event descriptions, combining stratified diversity sampling of real records with template-based augmentation to address vocabulary gaps and class imbalance.
5. Implement a training pipeline that fine-tunes a bert-base-cased model on the corpus, using focal loss with inverse-frequency class weighting, gradient clipping, learning-rate scheduling, and early stopping.
6. Curate a knowledge base of African armed groups, conflict-affected cities and regions, weapons, and the four-level taxonomy, and integrate it as a validation layer over raw NER output.
7. Implement a FastAPI service that exposes the trained model and knowledge base through routes for training management, inference, event storage, analytics, and knowledge base administration.
8. Implement a React-based front-end application that allows non-expert users to fine-tune models, run inference on documents, browse stored events, and review analytics.
9. Evaluate the system end to end on a held-out validation set and on representative real-world articles, reporting overall and per-entity metrics, latency, and qualitative observations from user acceptance testing.
10. Document limitations of the current implementation and recommend a concrete programme of future work, including hierarchical event classification at inference time and natural-language question answering against the event store.

1.5 Methods

The research will adopt a design-science methodology [13] coupled with empirical evaluation of the artefact. Each component will be built iteratively, with the lessons of one iteration informing the design of the next, and with quantitative evaluation against held-out data controlling for over-fitting at each stage. The principal methodical choices are summarised below; full detail will be provided in Chapter 4 and Chapter 5.

Annotation schema design. Drawing on the proposal taxonomy, the ACLED column conventions, and a pilot study of grounding rates in synthetically and naturally annotated text, the entity schema will be restricted to entity types that can be reliably found verbatim in source text. EVENT_TYPE and COUNTRY will be dropped from the schema because their

grounding rates will be shown by the pilot to be too low to support reliable supervision; instead, COUNTRY will be recovered deterministically from the knowledge base and the event-type hierarchy will be computed in a post-NER step.

Data acquisition and preparation. Raw event records will be obtained from ACLED via its open data exports. The records will be tokenised and labelled in BIO format using the column-to-entity mapping documented in Section 5.2. Stratified diversity sampling will then be applied to a target subset size of 35,000 records, with augmentation adding approximately 15,000 templated examples to expand vocabulary coverage of action verbs and victim-targeting constructions that are under-represented in raw ACLED text. The resulting 50,000-example corpus will be partitioned into 80 percent training and 20 percent validation.

Model fine-tuning. The bert-base-cased model from the Hugging Face hub will be fine-tuned for token classification with seventeen output labels (eight entity types times two BIO prefixes, plus the O label). Sub-word labels will be aligned by carrying the first-subword label to subsequent sub-words with the B-/I- transition handled explicitly. Training will use AdamW with linear warm-up, optional ReduceLROnPlateau scheduling, gradient clipping, and a focal-loss objective with inverse-frequency class weights computed from the training set.

Knowledge-base integration. A static knowledge base of African armed groups, conflict-affected cities, and the four-level taxonomy will be loaded at inference time. Raw entity spans will be validated against the knowledge base, with confidence scores adjusted upward for entities that match curated entries and downward for those that do not. The post-NER pipeline will then assemble a 5W1H record per detected event description.

System packaging. The trained model, the knowledge base, and the event store will be exposed through a FastAPI service with documented routes for each capability, and a React/TypeScript front-end will be built on top of it. The system will be containerised with Docker Compose for reproducible deployment.

Evaluation. The fine-tuned model will be evaluated against a held-out validation set using token-level accuracy, per-entity precision, recall, and F1, and through qualitative inspection on out-of-corpus news articles. End-to-end latency will be measured at inference time, and qualitative user acceptance testing will be conducted on a small group of representative users.

1.6 Scope and Limitations

Scope

The thesis addresses the extraction of 5W1H attributes from English-language news reports describing violent events in the African context. The artefact delivered comprises:

- A fine-tuned bert-base-cased model for token classification across the eight-entity BIO schema described above.

- A four-level hierarchical taxonomy of African violent events with approximately ninety-five terminal categories.
- A knowledge base of African armed groups, conflict-affected cities, and weapons.
- A FastAPI service exposing training, inference, event storage, analytics, and knowledge-base management.
- A React/TypeScript front-end allowing users to run training jobs, perform inference, manage events, and view analytics.
- A reproducible Docker-based deployment.

Out of Scope

The following are explicitly outside the scope of this thesis but are identified as priority future work in Chapter 7.

- Multilingual extraction. The model and pipeline target English text only. Arabic, French, Portuguese, Amharic, Swahili, and other African languages are out of scope.
- Real-time stream processing. Articles are processed on demand or in batch; the system does not target sub-second pipeline latency at scale.
- Predictive forecasting of future violence from extracted records.
- Automatic ingestion of news from third-party feeds. Articles are uploaded or submitted manually for inference.
- Natural-language question answering against the event store. Queries are formulated through structured filters in the analytics interface, not free-text questions.
- Hierarchical event classification by a learned classifier. The taxonomy is applied through deterministic post-NER rules informed by knowledge-base look-ups; a supervised hierarchical classifier is identified as future work.
- Production-grade hardening (high availability, multi-region replication, advanced authentication, audit logging) beyond a research prototype.

Limitations

The principal limitations of the work are the following.

1. **Data domain.** The training data is dominated by ACLED-derived event descriptions, supplemented with template-based augmentation. ACLED descriptions are concise and follow distinctive stylistic conventions; full-length news articles may diverge in style and may require additional fine-tuning data.
2. **Synthetic augmentation.** Approximately thirty percent of the training corpus is generated through templated augmentation. The templates were designed to mirror naturalistic constructions, but they remain narrower than the diversity of real reporting.

3. **English only.** The system processes English text. A substantial share of African conflict reporting is in French, Arabic, Portuguese, and various African languages and is therefore not covered.
4. **Single-pass NER.** The model performs flat NER over BIO labels. Nested entities (for example, a city name inside an organisation name) and overlapping spans are not modelled.
5. **Post-NER taxonomy.** Hierarchical taxonomic classification of each event is performed by rule-based post-processing using the knowledge base. A learned classifier would likely outperform the rule-based fallback for ambiguous events.
6. **Modest computational budget.** Training is conducted on a single workstation with Apple Silicon GPU acceleration. Larger model variants such as bert-large-cased or specialised long-context models were evaluated but not used as the production model.

1.7 Application of Results

The deliverables of this thesis have several immediate and prospective applications across the African early warning, humanitarian, security, and research ecosystems.

Operational early warning. AU-CEWS, regional economic communities, and national early warning centres can use the system to accelerate the conversion of news into structured event records, freeing analyst attention for interpretation and decision support. The hierarchical taxonomy supports multi-level queries from broad categories (“political violence in West Africa this quarter”) to specific event types (“suicide bombings attributed to JNIM in Mali”).

Humanitarian response. Humanitarian organisations require detailed information about violence affecting civilian populations and humanitarian access. The event store, with its actor, victim, location, and casualty fields, provides a base for protection analysis, needs assessment, and access mapping.

Security and defence analysis. Peace support operations and national security services can use the system to monitor non-state armed groups, track cross-border threats, and inform operational planning, recognising that automated output supplements rather than replaces classified intelligence streams.

Research. Academic researchers and policy institutes can use the annotated dataset, the taxonomy, and the extracted event corpus as inputs to quantitative conflict studies, comparative analyses, and methodological development in domain-specific NLP.

Capacity building. The open architecture, documented annotation schema, and reproducible deployment lower the technical barrier for African universities, research centres, and government bodies to adopt and adapt the system to their own monitoring needs.

The breadth of these applications underlines that even with the acknowledged limitations, the artefact produced by this thesis contributes value beyond its strict research contributions.

1.8 Organization of the Rest of the Thesis

The remainder of this thesis will be organised as follows. Chapter 2 will review relevant literature, covering the foundations of information extraction and named entity recognition, the transformer architecture and BERT, techniques for handling class imbalance in token classification, and the conflict-event database landscape that informs the taxonomy. Chapter 3 will review prior systems most directly related to this work, including violence-specific event extraction systems and the closest African-context predecessor, and will identify the specific gaps that VioNER bridges. Chapter 4 will present the proposed solution, including the high-level system architecture, the entity schema, the hierarchical taxonomy, the knowledge base, the training pipeline, the inference pipeline, and the web-application architecture. Chapter 5 will describe the implementation, including the technology stack, data preparation, model training, focal-loss objective, backend services, and frontend application. Chapter 6 will present the experimental results, including dataset statistics, training dynamics, overall and per-entity performance, ablation studies, error analysis, and user acceptance feedback. Chapter 7 will conclude the thesis with a summary of contributions, recommendations, and a prioritised future work programme that addresses the limitations identified in Section 1.6.

2. Literature Review

Five strands of literature inform this thesis: information and event extraction, named entity recognition, the transformer architecture and BERT, methods for handling severe class imbalance in token classification, and the conflict-event database tradition that frames the operational context. Each is summarised below, with a final section on knowledge-base and ontology approaches to event representation. Paper-by-paper reviews of the most directly related systems are deferred to Chapter 3.

2.1 Information Extraction and Event Extraction

The umbrella term Information Extraction (IE) [4] covers any technique that pulls structured records out of free text. The field is conventionally broken into four sub-tasks — named entity recognition, relation extraction, coreference resolution, and event extraction — and although the textbook arrangement is a pipeline, in practice a great deal of recent work bundles two or more of these into joint or end-to-end models. The architectural decision I make in §4.2 is to stay close to the textbook pipeline, on grounds of debuggability and clarity rather than peak accuracy.

Event Extraction is the sub-task I care about most directly. The operational definition I use is the one in Ahn [3]: an event is a verb or nominal predicate together with the participants

(agent, patient, instrument), the location, and the time that the predicate involves. The Automatic Content Extraction program [14] formalised this view in the early 2000s by publishing a typology of event types and their arguments, and that typology shaped the next decade of supervised event extraction. The Text Analysis Conference Knowledge Base Population track later picked up the baton with larger and more diverse evaluations.

The journalistic 5W1H frame [15] is a less rigid alternative. It sidesteps the question of which event types to enumerate by asking the same six questions — Who, What, Whom, Where, When, How — of every reported event. For news text this is a near-perfect fit: journalists are trained to answer those questions in roughly the first paragraph, and analysts read for exactly those slots. I adopt 5W1H, leave event-type classification to a post-NER step against the taxonomy in §4.4, and so avoid baking a fixed event-type inventory into the supervised learning problem itself.

Hogenboom and colleagues survey event extraction methods for decision support in two complementary papers [16], [4]. They identify three broad methodological families: data-driven (supervised machine learning on annotated corpora), knowledge-driven (linguistic and ontological rules), and hybrid approaches that combine the two. The hybrid family has dominated practical deployments because it allows domain knowledge to be encoded in rules where it is naturally expressible and statistical learning to be applied where rule authorship is intractable. The VioNER system presented in this thesis is hybrid in this sense: the entity extraction is data-driven, while the post-NER 5W1H structuring, the taxonomy classification, and the knowledge-base validation are rule-driven.

2.2 Named Entity Recognition

For the purposes of this thesis, NER means deciding for every token in a sentence whether it belongs to an entity I care about and, if so, which entity type. Before deep learning, the dominant approach combined hand-engineered features (capitalisation patterns, gazetteers, surrounding part-of-speech tags) with a sequence model that knew how to make a structured prediction over the whole sentence — Hidden Markov Models, Maximum Entropy Markov Models, and later Conditional Random Fields (CRFs) [17]. CRFs became the workhorse because they model how successive labels constrain each other, which matters when entities span multiple tokens and a B- must be followed by I-, not by another B-.

I encode entities in BIO, the simplest of the standard schemes: a B- prefix on the first token of an entity, an I- prefix on each continuation token, and an O label everywhere else. For a schema with k entity types, BIO produces $2k + 1$ labels — in my case, seventeen. Variants like BIOES or BILOU add explicit single-token and final-token labels to give the decoder more structure to lean on. I chose plain BIO for two reasons: the entity types in §4.3 are short enough that the extra labels would mostly be redundant, and seqeval-style evaluation tooling supports plain BIO without configuration.

The first wave of neural NER systems leant on word embeddings fed through bi-directional Long Short-Term Memory (BiLSTM) networks [18], typically combined with a character-

level convolutional or recurrent layer to handle out-of-vocabulary tokens, and a CRF head on top to keep the BIO transitions sane. Lample and colleagues' architecture [19] became the recipe most replicated for several years, and it showed that learned representations could replace much of what the feature-engineering era had hand-crafted.

The next jump came from contextual representations — ELMo [20] and particularly BERT [6]. Once a transformer encoder had absorbed general-purpose language structure during pre-training, swapping out the classifier head and fine-tuning the encoder produced results that surpassed every BiLSTM-CRF system on the CoNLL-2003 benchmark, and the recipe has stayed roughly the same since: contextual encoder, linear classifier, optional CRF head, cross-entropy or focal loss. VioNER follows this recipe.

The property that matters most for a low-resource setting like the African violent-event domain is that a transformer encoder absorbs enough general English during pre-training that a comparatively small specialised corpus can fine-tune it to good per-entity performance. The 50,000-example corpus I describe in Chapter 5 would be far too small for a from-scratch BiLSTM-CRF; it is sufficient for a fine-tuned BERT precisely because of this transfer.

Recent work has pushed for African-language NER resources directly. Adelani and colleagues' MasakhaNER [11] established a benchmark for ten African languages with annotations for PERSON, ORGANISATION, LOCATION, and DATE; their headline finding is that generic multilingual encoders such as mBERT and XLM-RoBERTa underperform on African text relative to encoders that have seen African corpora during pre-training. AfroLM [40] and AfroXLMR continued this line. My target is English-language reporting of African events rather than African-language reporting, so I use bert-base-cased, but the entity schema is strictly richer than MasakhaNER's four types, and an African-pre-trained backbone is a natural future swap (§7.5).

2.3 Transformer Models and BERT

The transformer [5] removed the recurrent step that had constrained sequence models for years and put self-attention in its place: each position in the input gets to look directly at every other position, through learned query / key / value projections, and the cost of that look-up is paid in parallel rather than in sequence. For my purposes the practical consequences matter more than the mathematical novelty: training can be parallelised on a GPU, long-range dependencies — exactly the kind that link an actor at the start of a sentence to a casualty count at the end — survive the encoding intact, and the architecture scales gracefully with parameter count and data.

BERT [6] is the encoder half of the original transformer turned into a representation learner. Its pre-training task is deliberately self-supervised: a randomly masked 15 percent of the input tokens get hidden, the model is asked to predict them from the surrounding context, and an auxiliary next-sentence-prediction head is trained alongside on pairs of sentences. After that pre-training run, the encoder carries enough generic English structure that I can

attach a small task-specific head and fine-tune the whole thing on a few tens of thousands of NER examples with sensible results.

For token classification specifically, the recipe is simple. BERT gives me one contextual vector per WordPiece token; I run a linear layer over each vector to produce logits across the seventeen-label vocabulary; I backpropagate a focal-loss objective with class weights (§2.4) through everything. The bert-base-cased checkpoint I use has twelve transformer layers, 768-dimensional hidden states, and roughly 110 million parameters. I use the cased variant deliberately: in English conflict reporting, capitalisation is extraordinarily informative for ACTOR and location entities, and the uncased variant throws that information away.

Two further properties of BERT influence the design of the present system. First, BERT uses WordPiece sub-word tokenisation; a single gold-labelled word may be split into multiple sub-word tokens, and the labels must be projected onto sub-words for training and recovered from sub-words at inference time. The convention adopted in this work, described in detail in Chapter 4, is to assign the original label to the first sub-word and to convert any leading B- prefix to I- for subsequent sub-words, while assigning -100 to special tokens so that they are ignored by the loss. Second, BERT has a maximum input length of 512 tokens; longer documents must be split into overlapping windows. This thesis processes ACLED event descriptions and most news article paragraphs without exceeding this limit, but window-based handling is implemented for completeness.

Several BERT variants and successors have been proposed. RoBERTa [21] improves on BERT by removing the next-sentence-prediction objective, training longer with larger batches, and using dynamic masking. DistilBERT [22] applies knowledge distillation to produce a smaller and faster model with modest accuracy loss. XLM-RoBERTa [23] extends the multilingual setting with strong performance on non-English text. The present work fine-tunes the canonical bert-base-cased because of the balance it strikes between accuracy and resource requirements; alternative backbones are evaluated as part of the ablation discussion in Chapter 6.

2.4 Class Imbalance in Token Classification

NER under BIO is, almost by construction, a deeply imbalanced classification problem. Most tokens in any natural-language sentence are not part of any entity, so they all carry the O label; across the corpus I assemble in Chapter 5, O accounts for roughly seventy-eight percent of all tokens. The remaining twenty-two percent is itself uneven: ACTOR, CITY, DATE, REGION, and DISTRICT are common enough to learn well, while VICTIM, ACTION, and CASUALTIES are the entities I most need to recover and the entities on which a naive learner does worst.

The literature gives three handles on this problem, and I take a position on each.

Re-sampling the training set [24] is the textbook first response. Oversample the sentences that contain a rare entity, undersample the ones full of O. The wrinkle in token

classification is that the oversampling decision lives at the example level — a single sentence — but the imbalance lives at the token level. Oversampling a sentence to recover one VICTIM token also pulls along thirty O tokens. My stratified diversity sampling (§5.3) is a compromise: it oversamples sentences that contain rare entity *types*, while selecting for entity-type diversity rather than entity-token count, so it grows the rare classes without pumping up O proportionally.

Class-weighted cross entropy [25] adjusts the loss rather than the sampler. Each class is given a weight, large for rare classes and small for the dominant ones, that multiplies its contribution to the gradient. Inverse-frequency weighting sets the weight to $1/f_c$ (optionally normalised); effective-number weighting [26] smooths this with a corpus-sample-overlap correction. I use inverse-frequency weights, computed once from the training-set distribution at the start of training, with a maximum-weight cap to prevent the rarest classes from dominating gradient updates.

Focal loss [12] takes a different angle. Originally proposed for dense object detection, where the foreground/background ratio is even worse than for NER, it down-weights examples the model is already confident about and concentrates the loss on the examples it is still uncertain on. The formulation, given a per-token cross entropy $CE(p, y) = -\log p_y$, is

$$FL(p, y) = -\alpha_y \cdot (1 - p_y)^\gamma \cdot \log p_y \quad (1)$$

where p_y is the model's predicted probability of the true class y , $\gamma \geq 0$ controls how aggressively easy examples are discounted (with $\gamma = 0$ recovering ordinary cross entropy), and α_y is an optional per-class weight that lets focal loss compose with class weighting. A growing body of token-classification work has reported gains from this combination over either ingredient alone.

The present work combines focal loss ($\gamma = 2$) with inverse-frequency class weighting, computes the weights from the training-set label distribution at the start of training, and excludes the special -100 label from both the loss and the weight computation. The implementation follows the formulation in equation (1), extended with optional label smoothing β for regularisation:

$$FL_{LS}(p, y) = -\alpha_y \cdot (1 - p_y)^\gamma \cdot \sum_c y'_c \cdot \log p_c \quad (2)$$

where the smoothed target distribution is defined by

$$y'_c = (1 - \beta) \cdot 1[c = y] + \beta / (C - 1) \cdot 1[c \neq y] \quad (3)$$

in which $1[\cdot]$ is the indicator function, C is the total number of classes, and $\beta \in [0, 1)$ is the smoothing factor ($\beta = 0$ recovers the one-hot target).

The inverse-frequency weight for class c is computed once at the start of training as

$$\alpha_c = T / (C \cdot \max(f_c, 1)) \quad (4)$$

where T is the total token count over the training set, C is the number of classes, and f_c is the count of class c . The maximum weight is clipped to ten to prevent gradient instability. This clipping also has the practical effect of treating the three rarest entities (VICTIM, ACTION, CASUALTIES) identically at the loss level even though their raw frequencies differ slightly.

In token classification the ignore index (-100) used for special tokens, sub-word continuations after the first sub-word of a word, and padding, must be respected by the loss. The custom focal loss in this work masks ignored positions before computing log-softmax, which both excludes them from the loss and keeps the per-position probability normalisation correct.

2.5 Evaluation Metrics for Named Entity Recognition

Two granularities of evaluation are conventionally reported for sequence-labelling NER: token-level and span-level (sometimes called entity-level).

Token-level metrics compare gold and predicted BIO labels at each token position. Accuracy is the share of positions where the labels agree, with special tokens excluded. Per-class precision and recall can also be computed at this granularity. Token-level accuracy is, however, a misleading headline metric for imbalanced tagging: with O constituting roughly seventy-eight percent of tokens in this work's corpus, a degenerate model that predicts O everywhere would already achieve seventy-eight percent accuracy.

Span-level metrics compare assembled entity spans rather than individual tokens. A predicted span (type t , start s , end e) is counted as a true positive if and only if a gold span exists with the same type and the same start and end positions. Partial overlaps count as a false positive (the prediction) and a false negative (the gold span). Per-entity precision is true positives divided by predicted spans; per-entity recall is true positives divided by gold spans; F1 is the harmonic mean. Macro F1 averages per-entity F1 with equal weight; micro F1 pools counts across entities before computing the single F1, weighting more populous entities more heavily.

The standard reference implementation of span-level NER evaluation is `seqeval` [39], which exposes both “default” (BIO with strict boundary matching) and “strict” variants. I report span-level precision, recall, and F1 with strict boundary matching to avoid inflated scores from partial credit, since downstream consumers of the extracted records (the knowledge-base validator, the event store, the analytics layer) treat boundary mismatches as distinct extractions rather than partial matches.

A subtle point arises with BIO encoding under sub-word tokenisation. The convention used here, with the first sub-word inheriting the B- or I- label of the underlying word and subsequent sub-words receiving an I- label converted from any leading B-, ensures that the recovered spans align with whole-word boundaries at inference time. Where this convention is not followed, span-level evaluation may be systematically optimistic or pessimistic depending on how sub-word predictions are merged.

2.6 Conflict Event Databases and Coding Schemes

The taxonomic structure of violent events is informed by a long line of conflict-event databases. Three are particularly influential.

ACLED. The Armed Conflict Location and Event Data Project [8] maintains a georeferenced database of political violence and protest events covering Africa, the Middle East, Latin America, South and South-East Asia, and other regions. Each ACLED record includes fields for event type, sub-event type, actors, location, fatalities, and a free-text note describing the event. ACLED's event type taxonomy distinguishes battles, explosions and remote violence, violence against civilians, protests, riots, and strategic developments, with sub-event types under each. ACLED is the principal data source for this thesis: the training corpus is constructed by labelling ACLED event notes against the column metadata, and the taxonomy presented in Annex B is influenced by ACLED's structure.

UCDP. The Uppsala Conflict Data Program [9] focuses on organised violence, distinguishing state-based conflict, non-state conflict, and one-sided violence, with fatality thresholds that exclude lower-intensity events. UCDP's careful definitional work, including the operational definition of organised actors and the distinction between battle-related and civilian deaths, informs the boundary between political violence and other categories in the taxonomy.

GDELT. The Global Database of Events, Language, and Tone [10] is an automatically generated event database using the CAMEO event coding framework [27]. CAMEO is a broad taxonomy covering both cooperative and conflictual events, organised in a four-digit hierarchical code. GDELT's scale is impressive, but its automated extraction is comparatively coarse and produces a high volume of events that may not satisfy the precision standard of operational early warning. CAMEO and GDELT are referenced in this thesis as illustrations of the scale-versus-precision trade-off and as sources of comparative taxonomy patterns.

The taxonomy of African violent events presented in this thesis is synthesised from ACLED and UCDP frames with additional categories that reflect African conflict dynamics, including pastoralist-farmer clashes, communal cattle raiding, and election violence. The taxonomy is presented in full in Annex B.

2.7 Knowledge Bases and Ontologies for Events

Knowledge bases (KBs) and ontologies complement extraction by providing reference data and semantic structure against which extracted items can be validated and reasoned over. Several event-oriented vocabularies have been proposed.

The Event Ontology by Raimond and colleagues defines an abstract notion of an event with participants, agents, places, and times. The Linking Open Descriptions of Events (LODE) vocabulary [28] simplifies the Event Ontology for use in linked data. Both vocabularies are expressed in RDF/OWL and emphasise interoperability across data sets.

For violence specifically, Piskorski and colleagues developed the Politically Motivated Violent Events (PMVE) ontology as the underpinning of the NEXUS system [29]. PMVE encodes violence-relevant classes and relations (perpetrator, victim, target, weapon, location, time) in OWL and supports rule-based reasoning over extracted facts. The taxonomy in this thesis is conceptually similar to PMVE in its breakdown of violence types, but is tailored to African conflict patterns and is implemented as a structured object in the application knowledge base rather than as an OWL ontology, on grounds of implementation simplicity and operational utility.

The knowledge base in VioNER is structured rather than ontological. It consists of normalised dictionaries of armed groups (with name variants, country of operation, region), African cities and regions (with country and parent administrative unit), and weapons (grouped by category). The KB is loaded at inference time and is used both to validate raw NER spans (a span tagged ACTOR is upweighted if it matches a known armed group) and to enrich extracted records (a CITY span is annotated with its country of operation when the city is present in the dictionary). The structured-knowledge-base approach sacrifices some expressivity relative to RDF/OWL but is markedly easier to maintain and to integrate with the application layer.

3. Related Work

Where Chapter 2 surveyed the field at the level of theory and methods, this chapter zooms in on the specific systems that VioNER either builds on or distinguishes itself from. Only concrete systems are reviewed — published papers, peer-reviewed conference proceedings, masters and doctoral theses, and technical reports — and each is read in light of what it tells me about the design choices I had to make in VioNER. A summary of the gaps that VioNER bridges closes the chapter.

3.1 General Event Extraction from News

Tanev, Atkinson, and Piskorski [1] built an early version of what operational crisis-monitoring infrastructure actually looks like at scale: a multilingual pipeline that pulls news from many feeds, recognises events through hand-written patterns, and clusters them geographically so that an emerging story can be tracked across sources. What I take from their work is the engineering attitude — they treat scale as the central problem rather than treating it as an afterthought. What I don't take is the pattern-based event recognition. Their system does not produce structured 5W1H records that a downstream analyst could query, and patterns alone do not scale across the actor-name and place-name diversity of African conflict reporting.

Hogenboom and colleagues survey the broader event-extraction landscape twice, with a slightly different cut each time [16], [4]. Their conclusion in both surveys is the same: data-driven methods generalise but lose precision on rare event types, knowledge-driven methods are precise but fragile, and the hybrid pipelines that mix the two outperform

either pure approach in operational settings. VioNER takes this seriously. The NER component is data-driven; the post-processing layer (5W1H grouping, KB lookup, taxonomy assignment) is rule-driven; the two communicate through confidence scores and KB metadata rather than trying to do the whole job in either paradigm.

YAGO [30] is a different beast — Suchanek, Kasneci, and Weikum extract millions of facts from Wikipedia infoboxes and WordNet hierarchies into a single OWL ontology. YAGO is not directly comparable to VioNER (its source is semi-structured rather than free text) but two of its design choices are instructive. First, where the source has structure, you should use it; an infobox is a gift compared to a paragraph. Second, embedding extracted facts in an ontology — not just storing them as rows — gives the result analytical leverage that flat records do not. The four-level taxonomy in §4.4 is my answer to that second observation, applied to a free-text extraction setting where I do not get YAGO’s infobox luxury.

Hienert and Luciano [31] extend the YAGO idea to historical events in Wikipedia using LODE. Their pipeline assumes the same semi-structured input. I cite it here mainly as a counterpoint: extracting from a Wikipedia article that already names “Battle of Adwa, 1896” in its title is a fundamentally easier problem than extracting events from ACLED notes or full-length news articles where the event has to be inferred from prose.

3.2 Violence-Specific Event Extraction Systems

The NEXUS system [29] is the closest stylistic precedent in violence-specific extraction. Piskorski, Tanev, and Wennerberg filter incoming news with keywords, run linguistic patterns to find violence-relevant sentences, then snap extracted entities onto the classes of their Politically Motivated Violent Events ontology. I draw two lessons from NEXUS. The principled ontology — PMVE — shows how much analytical reach a tight conceptual model gives downstream consumers, and my four-level taxonomy is partly a response to that. At the same time, NEXUS demonstrates how expensive pattern-based recognition becomes once the actor and place vocabularies shift: it was tuned to European security incidents, and porting it to the African continent — with several hundred armed groups and tens of thousands of locality names — would require essentially rebuilding the pattern base. I chose fine-tuned BERT precisely because it amortises that vocabulary work into the pre-training stage.

Two social-media-oriented papers sit at the edge of relevance to my work. Becker and colleagues [32] extract *planned* events by exploiting platform-specific structured fields (event titles, dates, locations posted on Eventful) alongside the unstructured discussion that surrounds them. Magnuson and colleagues [33] build a related Twitter recommendation system over Eventbrite. Both targets differ from violent events on every axis — events are unplanned and adversarial, the source is not authoritative, and there is no structured platform-side metadata to lean on — so I do not adopt their pipelines. The architectural lesson I do take is the explicit pairing of structured platform data with unstructured text; the knowledge base in §4.5 plays the role that Eventful or Eventbrite plays in Becker’s setting.

Aratefeh and Khreich [34] survey event detection on Twitter specifically and catalogue the supervised, unsupervised, and hybrid techniques that have been tried for the short-text, high-noise regime. VioNER processes news rather than tweets, so the survey does not drive my architecture, but the noise-handling and short-text techniques it catalogues are the natural starting point if I extend VioNER to citizen-journalist sources in future work.

3.3 Event Extraction in the African Context

The closest predecessor to VioNER is the work of Taye Abdulkadir Edris and Sungkur, who developed a system for extracting 5W characteristics of violent events in the African context [35]. Their system combines linguistic preprocessing with Stanford CoreNLP and machine-learning classification with Weka. They report results on a modest annotated corpus and demonstrate that domain-specific adaptation improves performance over generic baselines on African text.

The differences between Taye Abdulkadir's work and mine are substantial. First, that implementation uses Stanford CoreNLP and Weka, which limits access to modern transformer architectures; I use Hugging Face Transformers and a fine-tuned BERT. Second, the earlier annotated corpus is modest in size, while VioNER trains on approximately fifty thousand examples derived from ACLED and template augmentation. Third, this thesis develops an explicit hierarchical taxonomy of African violent events with approximately ninety-five terminal categories, curates a knowledge base of armed groups and locations, and packages the system as a deployable web application; Taye Abdulkadir's work stops at the model boundary. Fourth, I address class imbalance explicitly through focal loss and stratified sampling rather than leaving it to the supervised learner.

The relationship between the two works is therefore one of foundation and extension. Taye Abdulkadir established the feasibility of 5W extraction on African conflict text; this thesis extends that line in scale, architecture, taxonomy, knowledge-base integration, and operational packaging.

3.4 Hierarchical Event Classification

Hierarchical classification of events is most fully developed in the work of Wang and Zhao on Chinese news 5W1H semantic-element extraction [36]. They define the News Ontology Event Model (NOEM) and populate it with extracted assertions as RDF triples. While the linguistic challenges of Chinese differ from those of English, the architectural pattern of feeding extraction output into a structured ontology with hierarchical classes is a direct antecedent of the post-NER taxonomy pipeline in VioNER.

ACLED's own taxonomy, while operationally tuned, is also a precedent for hierarchical violence categorisation [8]. ACLED uses two levels (event type and sub-event type) and approximately twenty-five sub-event categories. The present work extends this in two directions: in depth, by adding two additional levels of granularity (an intermediate Level 2 and a detailed Level 4); and in breadth, by adding African-specific categories such as

pastoralist-farmer clashes and communal cattle raiding that are folded into other categories in ACLED.

A learned hierarchical classifier could in principle replace the rule-based taxonomy assignment used here. Methods for hierarchical classification range from top-down cascades, in which a separate classifier is trained at each level, to global classifiers that predict the entire path simultaneously [37]. Designing and training a learned hierarchical classifier is identified as future work in Chapter 7.

3.5 Summary of Gaps Addressed

Table 3.1 compares the most directly related prior systems with VioNER along seven dimensions: NLP backbone, entity-schema size, target languages, regional focus, taxonomy depth, presence of a curated knowledge base, and operational packaging.

Table 3.1: Comparative position of VioNER relative to prior systems

System / dimension	NLP backbone	Sc he ma	Lang uage s	Regional focus	Taxono my depth	KB integrati on	Operational packaging
NEXUS [29]	Pattern + linguistic	~8	Multi lingu al (EU)	Europe	2 levels	PMVE ontology	Internal prototype
Tanev et al. [1]	Pattern + linguistic	~6	Multi lingu al	Global	1 level	None curated	Internal system
YAGO [30]	Heuristic / rules	n/a	Engli sh	Global Wikipedi a	n/a	YAGO ontology	Knowledge base only
Wang & Zhao [36]	Pattern + semantic	5W 1H	Chin ese	Chinese news	NOEM ontology	NOEM	Research prototype
Taye Abdulkadir [35]	CoreNLP + Weka	5W	Engli sh	Africa	None	None	Model only
Adelani et al. (MasakhaNER) [11]	mBERT/XL M-R	4	10 Afric an	Africa	n/a	None	Benchmark + corpus
VioNER (this thesis)	bert-base-cased fine-tuned	8	Engli sh	Africa	4 levels (~95 categories)	150 groups + 200 cities + weapon	Full web app + API + Docker

System / dimension	NLP backbone	Sc he ma	Lang uage s	Regional focus	Taxono my depth	KB integrati on	Operational packaging
s							

The literature review and the related-work review converge on five gaps that this thesis addresses.

1. **Grounded entity schema.** Prior African-context work has either used very small schemas (person / organisation / location, as in MasakhaNER) or broader schemas that include entity types that cannot be reliably grounded in source text (the original VioNER proposal's 26-type schema). This thesis introduces a deliberately intermediate eight-entity schema in which every entity type has been verified to be grounded in source text in pilot evaluation (Section 1.5).
2. **Transformer-based NER for African violence.** Prior work in this domain has used pattern-based extraction (NEXUS, Tanev et al.) or pre-transformer machine learning (Taye Abdulkadir's CoreNLP + Weka). VioNER is, to the best of my knowledge, the first fine-tuned transformer NER model targeted specifically at African violent-event 5W1H extraction at the 50,000-example scale.
3. **African violent-event taxonomy at four levels.** To the best of my knowledge, no prior open taxonomy of African violent events at this level of granularity has been published. ACLED's taxonomy reaches two levels with approximately twenty-five sub-event categories; UCDP and GDELT/CAMEO are coarser-grained. The four-level VioNER taxonomy with approximately ninety-five terminal categories is presented in full in Annex B.
4. **Knowledge-base validation layer.** The combination of a curated knowledge base of African armed groups, cities, and weapons with a learned extractor — used for both validation (flagging geographically implausible extractions) and enrichment (canonicalising armed-group names, attaching country and region metadata to cities) — is, to the best of my knowledge, a contribution.
5. **Operational packaging.** Prior academic work in this area has stopped at the model boundary. NEXUS, Tanev et al., and Taye Abdulkadir all describe extraction systems without delivering a reproducible, documented web application that non-specialist users can operate end-to-end. This thesis delivers a full web application with documented APIs and a reproducible Docker Compose deployment.

4. The Proposed Solution

The design of VioNER answers a sequence of questions: what to extract, in what schema, how to train the extractor, how to make its output trustworthy, and how to put it in front of an analyst. Each section below addresses one of these in turn — design principles, system architecture, entity schema, hierarchical taxonomy, knowledge base, training pipeline,

inference pipeline, and the web application that sits on top. Implementation-level detail and specific technology choices are deferred to Chapter 5.

4.1 Design Principles

Six principles guide the design.

P1: Grounded supervision. Every entity type in the schema must be something that can be reliably found verbatim in source text. Entity types whose grounding rates were below an acceptable threshold during pilot study (specifically, `EVENT_TYPE` and `COUNTRY`) are excluded from the NER model and recovered downstream by deterministic post-processing against the knowledge base.

P2: Modular pipeline. The system is organised as a pipeline in which each stage has a well-defined input and output, with no hidden state. NER produces token labels; entity assembly produces spans; post-processing produces 5W1H records; knowledge-base validation produces enriched records. Each stage can be inspected, replaced, and unit-tested in isolation.

P3: Hybrid statistics and knowledge. A learned model handles generalisation over surface forms. A structured knowledge base handles deterministic look-ups (city-to-country resolution, armed group alias resolution, taxonomy classification). Each is used where it is strongest.

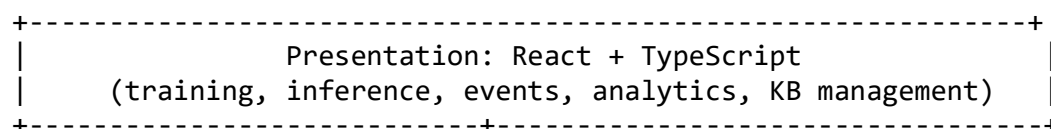
P4: Confidence is first-class. Every span produced by the NER component carries a confidence score derived from the softmax probabilities. Downstream stages can apply category-specific thresholds, and the UI surfaces confidence so that users see what they are trusting.

P5: Operational packaging. The model is exposed through a documented HTTP API and consumed by a web UI. Users do not need to know about checkpoints, tokenisers, or label vocabularies.

P6: Reproducibility. All datasets, training runs, and the final deployment are reproducible from documented scripts and configuration. Random seeds are fixed where applicable and recorded where not.

4.2 System Architecture

Figure 4.1 outlines the high-level architecture of VioNER. The system is organised in four layers: a model layer that hosts the trained BERT model, a service layer that exposes the model and supporting data stores through HTTP APIs, a data layer that persists events, training runs, and user accounts, and a presentation layer that provides the web UI.



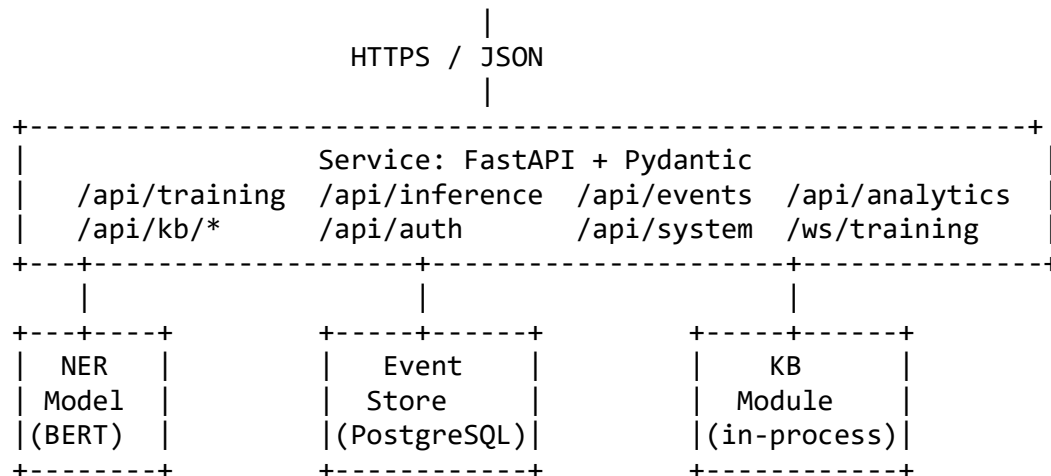


Figure 4.1: High-level architecture of the VioNER system

Figure 4.2 sketches the end-to-end processing pipeline that an incoming article traverses.

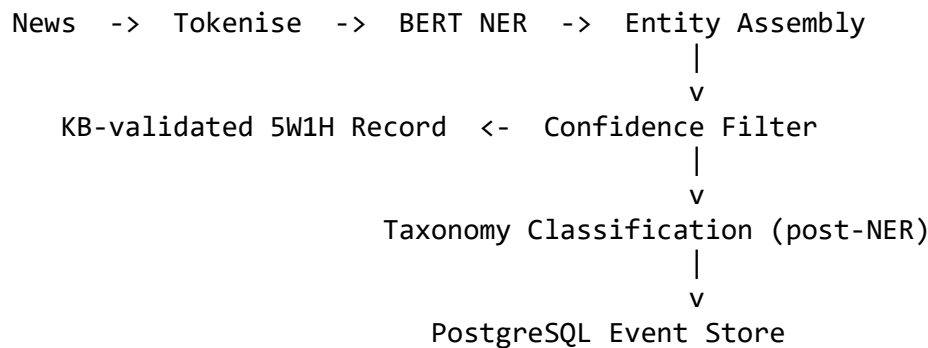


Figure 4.2: End-to-end processing pipeline from raw news text to queryable structured event records

Figure 4.3 expands the same pipeline as a sequence diagram in which each participating component is rendered as a vertical lane.

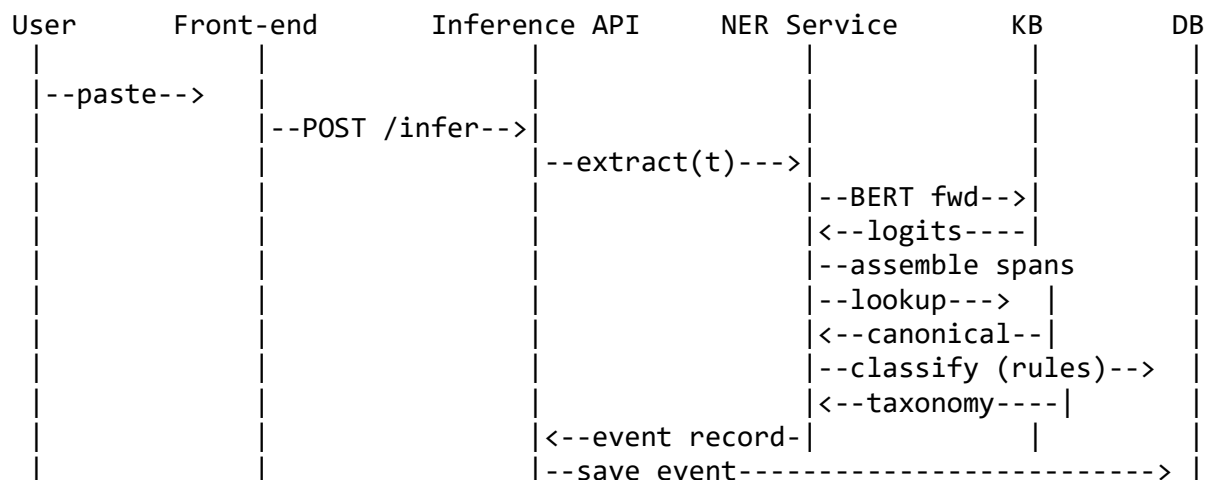




Figure 4.3: Sequence of calls during synchronous inference

The boundary between extraction (NER) and post-processing is deliberate: it allows the supervised learning problem to be cast narrowly, with the schema in 4.3, while still producing a richer output by combining the NER result with deterministic knowledge.

4.3 Entity Schema and BIO Encoding

The entity schema is the central design decision. As discussed under P1, the schema is restricted to entity types that pilot evaluation confirmed as reliably grounded in source text. The resulting eight entity types, organised under the 5W1H categories, are listed in Table 4.1.

Table 4.1: Eight-entity grounded schema for the VioNER NER component

5W1H category	Entity type	Description and examples
WHO	ACTOR	Armed groups, organisations, government forces (e.g., “Boko Haram”, “M23 rebels”, “Sudanese Armed Forces”, “gunmen”)
WHOM	VICTIM	Persons or groups affected (e.g., “civilians”, “villagers”, “protesters”, “traders”)
WHAT	ACTION	Verbs describing the event (e.g., “attacked”, “killed”, “ambushed”, “raided”)
WHEN	DATE	Temporal expressions (e.g., “on Monday”, “January 15, 2024”, “last week”)
WHERE	REGION	States, provinces, regions (e.g., “Borno State”, “Tigray”, “North Kivu”)
WHERE	CITY	Cities, towns, villages (e.g., “Maiduguri”, “Goma”, “Mogadishu”)
WHERE	DISTRICT	Districts, counties, localities (e.g., “Bama”, “Masisi”, “Lubero”)
HOW	CASUALTIES	Death/injury counts (e.g., “killed 12”, “3 dead”, “5 injured”)

EVENT_TYPE and COUNTRY are deliberately absent. EVENT_TYPE is recovered in the post-NER taxonomy step from the action verb, the actor type, and the surrounding context. COUNTRY is recovered from the knowledge base by looking up the most specific WHERE entity extracted from the text.

The schema is encoded in BIO format: for each entity type X, two labels B-X and I-X are introduced; together with the special O label for tokens outside any entity, the model has seventeen output classes. The BIO encoding allows the model to mark entity boundaries while remaining a flat sequence-labelling problem.

Figure 4.4 illustrates the encoding on a representative example.

Tokens:	"Al"	"Shabaab"	"fighters"	"killed"	"12"	"civilians"
	"in"	"Beledweyne"	"on"	"Sunday"		
Labels:	B-ACTOR	I-ACTOR	O	B-ACTION	B-CASUALTIES	B-VICTIM
	O	B-CITY	O	B-DATE		

Figure 4.4: BIO encoding example for a sentence describing a violent event. Multi-token entities such as “Al Shabaab” are encoded by a leading B- tag followed by I- tags

Sub-word tokenisation introduces a complication: a single gold-labelled word may be split into several sub-word tokens, and the labels must be projected onto the sub-words during training. The convention adopted here, summarised as Algorithm 4.1 in Section 4.6, assigns the original label to the first sub-word and converts any leading B- prefix to I- for subsequent sub-words, while assigning the ignore index (-100) to special tokens such as [CLS], [SEP], and [PAD] so that they are excluded from the loss.

4.4 Hierarchical Violent Event Taxonomy

The taxonomy I developed for this thesis is a four-level hierarchy. Level 1 names four broad categories of violence; Level 2 introduces eighteen intermediate types nested under them; Level 3 refines those into roughly fifty specific event types; and Level 4 adds another twenty or so detailed subtypes where the operational distinctions warrant it. The synthesis draws on ACLED’s primary event types [8], UCDP’s distinction between state-based, non-state, and one-sided violence [9], and the violence-relevant classes of PMVE [29], with African-specific extensions for pastoralist / farmer clashes and communal cattle raiding that none of those frameworks cover at this depth. The complete tree is in Annex B; the present section summarises Levels 1 and 2 in Tables 4.2 and 4.3 and gives the visual outline in Figure 4.5.

Table 4.2: Level 1 categories of the hierarchical taxonomy

Level 1	Working definition adopted in this thesis
Political Violence	Acts intended to contest state authority, push political change, or advance an ideological programme; perpetrators range from rebel and insurgent groups to terrorist organisations and political factions.
Criminal Violence	Acts whose primary motivation is economic gain or the territorial control of a criminal enterprise; perpetrators are gangs, bandits, organised-crime networks, or individual criminals.

Level 1	Working definition adopted in this thesis
Communal Violence	Acts between identity-defined groups — ethnic, religious, clan, or pastoralist / farmer communities — over land, water, livestock, or social standing, where the state is not the primary target.
State Violence Against Civilians	Lethal or severely coercive acts by police, military, or paramilitary forces directed at non-combatants outside an active armed-conflict context, encompassing extrajudicial killing, protest crackdowns, and forced displacement.

Table 4.3: Level 2 intermediate violence types

Level 1	Level 2
Political Violence	Rebellion/Armed Insurgency; Terrorism; Coup and Regime Change Violence; Election Violence; Political Repression.
Criminal Violence	Organised Crime Violence; Armed Robbery/Banditry; Kidnapping for Ransom; Criminal Gang Violence.
Communal Violence	Ethnic/Tribal Conflict; Religious Violence; Resource-Based Conflict; Pastoralist-Farmer Clashes.
State Violence Against Civilians	Extrajudicial Killings; State Repression of Protests; Mass Atrocities by State Forces; Forced Displacement by State; Arbitrary Detention with Violence.

VIOLENT EVENTS TAXONOMY

-- POLITICAL VIOLENCE	
-- Rebellion / Armed Insurgency	(Armed Clash, Ambush, Rebel Attack, Forced Recruitment)
-- Terrorism	(Bombing, Armed Assault, Hostage-Taking, Assassination, Soft Targets)
-- Coup and Regime Change	(Military Coup, Coup-Related Violence, Assassination)
-- Election Violence	(Campaign Violence, Voting Day Violence, Post-Election Violence)
-- Political Repression	(Protest Suppression, Targeted Killings, Mass Arrests)
-- CRIMINAL VIOLENCE	
-- Organised Crime Violence	(Gang Warfare, Assassination, Violence Against Law Enforcement)
-- Armed Robbery / Banditry	(Highway, Bank/Business, Home Invasion, Cattle Raiding)
-- Kidnapping for Ransom	(Individual/Family, Maritime/Piracy)
-- Criminal Gang Violence	
-- COMMUNAL VIOLENCE	

	-- Ethnic/Tribal Conflict	(Ethnic Clash, Ethnic Massacre, Ethnic
	Revenge Attack)	
	-- Religious Violence	(Sectarian Violence, Attack on Religiou
	s Community, Site Desecration)	
	-- Resource-Based Conflict	(Land, Water, Mining/Resource Extractio
	n)	
	`-- Pastoralist-Farmer Clashes	(Grazing, Cattle Raiding (Communal), Re
	venge Raid)	
`--	STATE VIOLENCE AGAINST CIVILIANS	
	-- Extrajudicial Killings	(Summary Execution, Enforced Disappeara
	nce, Torture Death)	
	-- State Repression of Protests	(Shooting of Protesters, Violent Disper
	sal Resulting in Deaths)	
	-- Mass Atrocities	(Massacre by State Forces, Ethnic Clea
	nsing by State)	
	-- Forced Displacement	(Violent Eviction, Village Burning)
	`-- Arbitrary Detention	(Violent Mass Arrest)

Figure 4.5: Four-level taxonomy hierarchy (visual outline of Levels 1 to 3)

Each terminal category in the taxonomy is documented with a definition, classification criteria, distinguishing features, typical keywords used in news reports, and worked examples; the full documentation is provided in Annex B. Decision rules for ambiguous cases (for example, criminal versus terrorist kidnapping, or state violence against protesters versus political repression of protests) are also documented in Annex B.

Within VioNER, the taxonomy is applied as a post-NER step. Given an extracted set of entities and the surrounding text, the taxonomy classifier inspects the actor type, the action verb, contextual cues (target type, weapons, casualties), and looks up the actor in the knowledge base to identify its known classification. The result is a Level 1 to Level 3 (and where appropriate Level 4) classification of the event. A learned hierarchical classifier could replace this rule-based step in future work.

4.5 Knowledge Base Design

The knowledge base is a structured, in-process resource loaded at service startup. It consists of three dictionaries.

Armed Groups. Approximately 150 entries cover major armed groups active in Africa, each represented by a canonical name, a list of aliases, a country of operation, a region, and a group type (militia, terrorist, rebel, or government). Entries are organised by region (East, West, North, Southern, Central Africa) for readability. Example entries include Al-Shabaab (Somalia), Boko Haram (Nigeria), M23 (DRC), Rapid Support Forces (Sudan), JNIM (Mali), and Wagner Group (multiple). The full list is in Annex C.

Cities and Regions. Approximately 200 conflict-affected cities are recorded with their country and parent administrative region. The dictionary also covers all 54 African

countries with their capitals and major regions. Examples include Maiduguri (Nigeria, Borno), Goma (DRC, North Kivu), Mogadishu (Somalia, Banaadir), El Fasher (Sudan, North Darfur), and Bamako (Mali, Bamako Capital District). The dictionary is used both to resolve city-country ambiguity at inference time and to flag geographically implausible extractions.

Weapons and Tactics. A categorised list of weapons (firearms, explosives, edged weapons, fire/arson, heavy weapons) and tactical methods (ambush, raid, mass shooting, suicide bombing). The list is used by the post-NER taxonomy classifier to inform Level 3 and Level 4 classification.

Table 4.4 summarises the size of each knowledge-base resource. The knowledge base is also used by the validator component to attach metadata to extracted entities. For an ACTOR span that matches a known armed group alias, the validator attaches the canonical name, country, region, and group type. For a CITY span, it attaches the country and region. For a weapon mention, it attaches the category. Mismatches (for example, an actor whose known country of operation disagrees with the location extracted in the same sentence) lower the event’s overall confidence score and are surfaced in the analytics view for manual review.

Table 4.4: Knowledge base content summary

Resource	Approximate size
Armed groups	150
Cities (conflict zones)	200
African countries	54
Weapon types/categories	30
Taxonomy categories	95

4.6 Training Pipeline

The training pipeline turns raw ACLED event records into a fine-tuned BERT model. Its main stages are preprocessing, sampling and augmentation, sub-word label alignment, loss configuration, and checkpointed training with early stopping.

Preprocessing

The raw input is a JSONL file of ACLED event records with fields including event_id, event_date, notes (the free-text description), fatalities, actor1, location, and admin1. The preprocessing module tokenises the notes field on whitespace and punctuation, then projects the structured columns onto BIO labels by mapping each column to its entity type and locating its value within the tokenised notes. Tokens that do not match any column value are labelled O. The output is a JSONL file with one record per event, each containing tokens, labels, text, and entities fields.

Stratified diversity sampling

The raw corpus is large (approximately 212,000 events) but diversity-poor: many events share repeating phrasing. The `create_training_subset.py` script (Algorithm 4.2) selects a 35,000-example subset that maximises diversity while ensuring that rare entity types receive proportionate representation. The algorithm operates in three phases: first, it selects examples that contain at least one rare entity (VICTIM, ACTION, CASUALTIES) until the rare-class budget is met; second, it selects examples with high entity diversity (number of distinct entity types); third, it fills the remaining budget by random sampling from the residual pool.

Augmentation

Template-based augmentation adds approximately 15,000 synthetic examples to expand vocabulary coverage of action verbs and victim constructions. Templates are populated with knowledge-base entries (armed groups, cities, regions) and a curated lexicon of action verbs grouped into location-taking, victim-taking, and clash categories. The full template catalogue is in Annex E.

The combined 50,000-example corpus is partitioned 80/20 into training (40,000) and validation (10,000) splits, preserving stratification across entity types.

Sub-word label alignment

Algorithm 4.1 describes the sub-word alignment used to project word-level labels onto BERT WordPiece tokens.

```
-----
Input:  tokens t[1..n], labels y[1..n], tokenizer T
Output: aligned_label_ids l[1..m] with m >= n
-----

enc      <- T(tokens = t, is_split_into_words = True)
word_ids <- enc.word_ids()
l        <- empty list
prev     <- null
for each w in word_ids do
  if w is null then
    append IGNORE_INDEX (-100) to l
  else if w != prev then
    append label2id[ y[w] ] to l                # first sub-word
  else                                          # later sub-word
    label <- y[w]
    if label starts with "B-" then
      label <- "I-" + label[2:]                # B->I transition
    end if
    append label2id[ label ] to l
  end if
  prev <- w
end for
```

```
return 1
```

Algorithm 4.1: Sub-word label alignment for BIO tagging

Training hyperparameters

The principal hyperparameters are listed in Table 4.5.

Table 4.5: Training hyperparameters

Hyperparameter	Value
Pre-trained model	bert-base-cased
Number of labels	17
Maximum sequence length	512
Batch size	16
Gradient accumulation	2
Learning rate	2×10^{-5}
Optimiser	AdamW
Weight decay	0.01
Warmup steps	500
Maximum epochs	10
LR scheduler	ReduceLROnPlateau (factor 0.5, patience 2) with warmup
Early stopping	patience 5, threshold 0.001
Loss function	Focal loss ($\gamma=2.0$) with inverse-frequency class weights
Label smoothing	0.1
Gradient clipping	1.0
Device	Apple Silicon (MPS) when available, otherwise CUDA or CPU

Focal loss with class weighting

The custom loss combines focal loss with inverse-frequency class weighting and optional label smoothing. Algorithm 4.4 summarises its behaviour.

Input: logits $z[1..N, 1..C]$, targets $y[1..N]$, class weights $\alpha[1..C]$,
focusing parameter γ , ignore index I , label smoothing β
Output: scalar loss L

```
mask[n]  <- (y[n] != I)                for n = 1..N
```

```

N_valid  <- sum(mask)
log_p    <- log_softmax(z, dim = -1)  # shape N x C
if  $\beta > 0$  then
  y_smooth[n,c] <- (1 -  $\beta$ )·1[c = y[n]] +  $\beta/(C - 1)$ ·1[c != y[n]]
  CE[n]        <- -  $\sum_c$  y_smooth[n,c] · log_p[n,c]
else
  CE[n] <- - log_p[n, y[n]]
end if
p_true[n]  <- exp( log_p[n, y[n]] )
modulator[n] <- (1 - p_true[n]) $^\gamma$ 
weight[n]  <-  $\alpha$ [ y[n] ]
L_n[n]     <- mask[n] · weight[n] · modulator[n] · CE[n]
return (  $\sum_n$  L_n[n] ) / max( N_valid, 1 )
-----

```

Algorithm 4.4: Focal loss with inverse-frequency class weighting

The class weights are computed from the training-set distribution at the start of training. The weight for class c is $w_c = T / (C * f_c)$, where T is the total token count, C is the number of classes, and f_c is the count of class c . The O class receives a low weight; rare entity classes receive high weights.

Checkpointing and early stopping

After every epoch, the model and tokenizer are saved to `models/{model_name}_{timestamp}/epoch_{NN}/`. Whenever the validation loss improves by more than the early-stopping threshold, the checkpoint is also copied to `best/`. A `training_config.json` file at the root of the run records the current epoch, the best epoch, the best validation loss, and a flag indicating whether training has completed. This structure supports resumption from any epoch with the `--resume` flag and extension of completed runs with the `--extend-epochs` flag.

4.7 Inference and Post-Processing

The inference pipeline transforms a raw input text into a structured 5W1H record.

Algorithm 4.5 describes the steps.

```

-----
Input:  text x, NER service M, knowledge base K,
        per-category thresholds  $\tau[c]$ 
Output: structured event record R
-----
(tok, offsets) <- tokenize(x)
logits         <- M.forward(tok)
pred[i]        <- argmax_c logits[i, c]           for all i
probs[i]       <- softmax(logits[i, ·])           for all i
spans          <- empty list
i <- 1
while i <= |tok| do

```



```

if id2label[pred[i]] starts with "B-" then
  start <- offsets[i].start
  t      <- label[2:]
  confs  <- [ probs[i, pred[i]] ]
  j      <- i + 1
  while j <= |tok| and id2label[pred[j]] == "I-" + t do
    confs <- confs ++ [ probs[j, pred[j]] ]
    j      <- j + 1
  end while
  end_off <- offsets[j - 1].end
  spans  <- spans ++ [ (t, start, end_off, mean(confs)) ]
  i      <- j
else
  i <- i + 1
end if
end while
spans_filt <- { s in spans : confidence(s) >=  $\tau$ [ category_of(s) ] }
R.entities <- spans_filt
R.who, ..., R.how <- group_by_5w1h(spans_filt)
for each s in R.who do
  R.who_meta[s] <- K.armed_groups.lookup(s.surface)
end for
for each s in R.where do
  R.where_meta[s] <- K.locations.lookup(s.surface)
end for
R.taxonomy <- classify_taxonomy(R, K)
R.confidence <- aggregate_confidence(spans_filt, R.taxonomy)
return R
-----

```

Algorithm 4.5: Post-NER 5W1H structuring with knowledge-base validation

Confidence thresholds are calibrated by category: WHO 0.70, WHOM 0.70, WHAT 0.60, WHEN 0.80, WHERE 0.70, HOW 0.75. These thresholds were determined by inspection of validation-set behaviour: WHEN demands high precision because dates are easy to verify and false positives are highly visible; WHAT is allowed to be more permissive because actions are typically supported by surrounding context.

Multi-event texts are handled by a segmentation step that splits long documents into event-bearing sentences before running NER. The segmentation module (pipeline/segmentation.py) uses sentence boundary detection and a small set of indicator phrases (for example, “in a separate incident”, “earlier that week”) to identify event boundaries.

4.8 Web Application Architecture

The web application is organised as a single-page React/TypeScript front-end backed by a FastAPI service. The front-end communicates with the back-end exclusively through JSON over HTTPS, with a single WebSocket channel reserved for streaming training progress.

The back-end exposes the following route groups, all under `/api`:

- `/auth/*` for authentication and demo-user provisioning.
- `/training/*` for starting, monitoring, listing, and managing training runs; sub-routes cover checkpoint management, training data inspection, and post-training evaluation.
- `/inference/*` for synchronous inference on a single text or document.
- `/events/*` for storing, querying, and updating extracted events.
- `/analytics/*` for aggregated views (events per region, per actor, per time period).
- `/kb/actors`, `/kb/locations`, `/kb/taxonomies` for managing the knowledge base resources.
- `/system/*` for service health, version information, and configuration introspection.
- `/history/*` for recording per-user activity (recent inferences, saved queries).
- `/ws/training/{session_id}` as the WebSocket channel for live training progress.

The front-end mirrors these routes with a screen for each. Section 5.7 describes the implementation in detail; screenshots are included in Annex D.

5. Implementation

Where Chapter 4 described what VioNER does and why, this chapter documents how it actually does it: the technology stack, the data preparation flow, the training procedure, the focal-loss implementation, the back-end services and API surface, the front-end application, and the containerised deployment. Code listings are kept short throughout; the full source lives in the accompanying repository.

5.1 Technology Stack

The system is implemented in Python 3.11 for the back end and machine learning components, and in TypeScript with React 19 for the front end. PostgreSQL serves as the persistent data store. Docker Compose orchestrates the development environment. Table 5.1 and Table 5.2 list the back-end and front-end stacks respectively.

Table 5.1: Back-end technology stack

Component	Choice
Language	Python 3.11
Web framework	FastAPI 0.115
ASGI server	Uvicorn
Validation	Pydantic v2
ML framework	PyTorch 2.6
Pre-trained models	Hugging Face Transformers 4.46

Component	Choice
Tokenisers	Hugging Face Tokenizers (WordPiece)
ORM	SQLAlchemy 2.0
Database driver	psycopg2
Migration	Alembic (planned, currently DDL via SQLAlchemy)
WebSocket	fastapi.WebSocket
Authentication	JWT (PyJWT)
Testing	pytest
Logging	Python logging with structured output
Containerisation	Docker, Docker Compose

Table 5.2: Front-end technology stack

Component	Choice
Language	TypeScript 5.6
UI framework	React 19
Routing	React Router 7
Build tooling	Vite
Styling	TailwindCSS + shadcn/ui components
Data fetching	Native fetch with custom service wrappers
State management	React context and useReducer per feature
Charting	recharts
Tables and grids	TanStack Table
Type checking	tsc + react-router typegen

The motivation for these choices is summarised below. FastAPI gives type-driven API definition with automatic OpenAPI generation and strong asynchronous support; the OpenAPI document is consumed by the front-end to keep type definitions aligned. PyTorch and Hugging Face Transformers are the de-facto standard for transformer fine-tuning. SQLAlchemy provides a stable ORM. React 19 with React Router 7 gives a modern routing surface (including file-based route definitions) and fast iteration during development. TailwindCSS combined with shadcn/ui gives a coherent component library without locking the project into a heavy design system. Vite provides fast development builds and production bundling.

5.2 Data Acquisition and Preprocessing

The primary data source is the ACLED open dataset, accessed through the ACLED API. The retrieved records cover the 54 African countries and span multiple years. The raw extract is approximately 212,590 event records.

The preprocessing module `pipeline/preprocessing.py` converts the raw extract into BIO-tagged training data. The principal steps are:

1. **Loading.** Each event is loaded from JSONL, with required fields verified and missing values defaulted.
2. **Tokenisation.** The free-text notes field is tokenised using a regex-based splitter that preserves punctuation and quotation marks but separates them from adjacent tokens.
3. **Column-to-entity projection.** Each structured column (e.g., `actor1`, `location`, `admin1`, `fatalities`) is mapped to its entity type per the table in `pipeline/config.py` (see Annex A). The corresponding value is located within the tokenised notes via case-insensitive substring matching; matched token positions are tagged with the appropriate BIO label.
4. **Output.** Each event is written to JSONL with tokens, labels, text, and entities fields. Train/validation split is 80/20.

The pre-processed corpus consists of:

- Train: 170,072 examples
- Validation: 42,518 examples
- Total: 212,590 examples

This is the upstream pool from which the diverse subset is sampled in Section 5.3.

5.3 Stratified Sampling and Augmentation

The `create_training_subset.py` script implements the stratified diversity sampling described in Section 4.6 and described in Algorithm 4.2.

```
-----
Input:  Pool P, target size T, rare-entity budget R,
        diversity budget D, rare set  $\Sigma = \{\text{VICTIM, ACTION, CASUALTIES}\}$ 
Output: Selected subset S with  $|S| = T$ 
-----
for each p in P do
    rare_score[p] <- | { e  $\in$  entities(p) : type(e)  $\in$   $\Sigma$  } |
    div_score[p]  <- | distinct( type(e) for e in entities(p) ) |
end for
P_rare <- top-R items of P ordered by rare_score desc
P_div  <- top-D items of ( P \ P_rare ) ordered by div_score desc
P_rest <- random-sample( P \ (P_rare  $\cup$  P_div), T - R - D )
S      <- P_rare  $\cup$  P_div  $\cup$  P_rest
return S
-----
```

Algorithm 4.2: Stratified diversity sampling for entity coverage

In the production run, $T = 35,000$, $R = 12,000$, $D = 11,666$, and the random remainder is 11,334.

Augmentation is implemented in `scripts/augment_training_data.py` following Algorithm 4.3.

```
-----
Input:  KB groups G, locations L, weapons W, verb lexicons V,
        template list T, augmentation budget N
Output: Synthetic example list A with  $|A| = N$ 
-----
A <- empty list
while  $|A| < N$  do
    tmp1      <- choose_random(T)
    actor     <- sample compatible group from G for tmp1
    loc       <- sample location from L matching actor.country
    verb      <- sample verb from V[ type(tmp1) ]
    n_k, n_i  <- sample plausible casualty counts
    text      <- render(tmp1, actor, loc, verb, n_k, n_i)
    (tok, lbl) <- tokenize_and_bio_label(text, slots)
    A         <- A ++ [ { tokens: tok, labels: lbl, source: "aug" } ]
end while
return A
-----
```

Algorithm 4.3: Template-based augmentation

The augmentation budget is 15,000 examples. Together with the 35,000 sampled examples, the final training corpus is 50,000 examples, partitioned 40,000 / 10,000 into train and validation.

Sample verb lexicons (location-taking verbs include “attacked”, “raided”, “stormed”, “bombed”, “shelled”; victim-taking verbs include “killed”, “wounded”, “abducted”, “displaced”; clash verbs include “clashed with”, “exchanged fire with”, “battled”) together with the template catalogue are reproduced in Annex E.

5.4 Model Training Implementation

The trainer is implemented as `pipeline/training.py`, organised around a `ViolentEventNER` class that wraps the Hugging Face model, the tokenizer, the dataset, and the training loop.

The constructor receives a `ModelConfig` dataclass that captures all hyperparameters (Table 4.5). Device selection prefers Apple Silicon GPU (MPS) when available, falls back to CUDA where present, and ultimately to CPU. The `load_model` method instantiates `AutoTokenizer.from_pretrained` and `AutoModelForTokenClassification.from_pretrained` for `bert-base-cased`, configures the model with the 17-label classification head, and moves it to the selected device.

The `NERDataset` class implements `torch.utils.data.Dataset`. Its `__getitem__` runs the tokenizer with `is_split_into_words=True`, retrieves `word_ids`, and applies the alignment in Algorithm 4.1. Special tokens are assigned label -100, which PyTorch ignores in the default cross-entropy loss and which the custom focal loss also respects.

The training loop iterates `num_epochs` epochs, with a per-epoch forward/backward pass for training and a no-grad validation pass. Gradient clipping bounds the L2 norm of the gradient at 1.0. The linear warm-up schedule is applied during the first `warmup_steps` optimisation steps, after which the `ReduceLROnPlateau` scheduler takes over and reduces the learning rate by a factor of 0.5 when the validation loss fails to improve for two consecutive epochs.

Early stopping is implemented by tracking the best validation loss seen so far and a counter of epochs without improvement. When the counter reaches the patience threshold, training stops. The best model is restored from the best/ checkpoint after termination.

Resumption is supported by `resume_training`, which reads the saved `training_config.json`, locates the most recent epoch folder, and re-enters the training loop at the appropriate epoch. Extension of a completed run is supported by the `--extend-epochs` flag, which increments the total epoch count and restarts the loop at the next epoch.

5.5 Focal Loss and Class Weighting

The custom loss is implemented in `pipeline/loss.py`. The `FocalLoss` class extends `torch.nn.Module` and implements Algorithm 4.4. The principal points of the implementation are:

- Logits and targets are flattened to two-dimensional and one-dimensional tensors respectively before the loss is computed.
- The ignore mask (`target == -100`) is applied before any softmax computation to keep the loss numerically stable.
- Label smoothing, when enabled, distributes mass across the vocabulary; the focal modulating factor remains applied to the smoothed distribution.
- The per-class weights are passed in as a 1D tensor on the same device as the logits; they are picked up by indexing.

The `compute_class_weights` function in the same module computes inverse-frequency weights from a label-count dictionary:

$$w_c = T / (C * \max(f_c, 1))$$

with optional normalisation. The training script logs the resulting O weight, the maximum weight, and the minimum weight to make the weighting transparent. The training-set label distribution and the resulting class weights are reproduced in Annex E.

The `ClassWeightedCrossEntropy` class provides a non-focal weighted alternative for ablation.

5.6 Backend Services and API

The back-end is implemented in `backend/main.py` with route handlers organised under `backend/api/`. Figure 5.1 sketches the module organisation.

<code>backend/</code>	
<code>main.py</code>	FastAPI app, lifespan, CORS, error handler
<code>config.py</code>	Settings (Pydantic BaseSettings)
<code>api/</code>	
<code>auth/router.py</code>	login, register, demo user provisioning
<code>training/</code>	
<code>router.py</code>	start/stop/list training jobs
<code>checkpoint.py</code>	list, load, delete checkpoints
<code>data.py</code>	upload and inspect training data
<code>evaluation.py</code>	post-training evaluation
<code>inference/router.py</code>	/api/inference endpoints
<code>events/router.py</code>	/api/events CRUD and search
<code>analytics/router.py</code>	/api/analytics aggregations
<code>kb/</code>	
<code>actors.py</code>	/api/kb/actors
<code>locations.py</code>	/api/kb/locations
<code>taxonomies.py</code>	/api/kb/taxonomies
<code>system/router.py</code>	/api/system health, version, config
<code>history/router.py</code>	/api/history recent activity
<code>websocket.py</code>	/ws/training streaming progress
<code>services/</code>	
<code>ner.py</code>	NERService (model loading, inference, 5W1H)
<code>training.py</code>	TrainingService (async job manager)
<code>evaluation.py</code>	metric computation
<code>pipeline/</code>	
<code>training.py</code>	ViolentEventNER (Section 5.4)
<code>loss.py</code>	FocalLoss, ClassWeightedCrossEntropy
<code>config.py</code>	LabelConfigs, ModelConfig
<code>segmentation.py</code>	multi-event splitting
<code>kb.py</code>	KB data and lookups
<code>validator.py</code>	NER output validation against KB
<code>database/</code>	
<code>connection.py</code>	engine creation, session management
<code>models/</code>	SQLAlchemy ORM models
<code>init/</code>	DDL bootstrap

Figure 5.1: Back-end module organisation

The application lifespan handler initialises the database connection, loads the NER model from the configured checkpoint path, and starts the training service. Failure to load the

model degrades the service to inference-disabled state but does not prevent startup; the front-end then surfaces a banner explaining that inference is temporarily unavailable.

Pydantic models defined under each router specify request and response shapes. FastAPI automatically generates the OpenAPI schema at /docs (Swagger UI) and /redoc. The schema is exported during the front-end build to keep TypeScript types in sync.

The `NERService.extract` method (`services/ner.py`) implements the algorithm from Section 4.7: tokenisation, BERT inference, entity-span assembly, confidence-based filtering, 5W1H structuring, and KB enrichment. The method returns a JSON-serialisable dictionary that the inference route returns directly.

The `TrainingService` (`services/training.py`) manages training jobs asynchronously. A new job spawns a subprocess that runs `pipeline/training.py` with the supplied arguments. The service listens for log lines on the subprocess's standard output and forwards them over the per-session WebSocket. Cancellation is supported by signalling the subprocess and writing a final state record to PostgreSQL.

PostgreSQL stores users, events, training runs, and inference history. The schema is reproduced in Annex D.

5.7 Frontend Application

The front-end is implemented as a Vite-built React 19 single-page application in `frontend/`. It uses React Router 7 file-based routing, with the route map shown in Figure 5.2.

<code>frontend/src/</code>	
<code> root.tsx</code>	app shell, error boundary
<code> routes/</code>	
<code>index.tsx</code>	landing
<code>auth/login.tsx</code>	
<code>inference/index.tsx</code>	run inference on text
<code>inference/upload.tsx</code>	upload a document
<code>training/index.tsx</code>	list/start runs
<code>training/\$id.tsx</code>	training run detail with live progress
<code>events/index.tsx</code>	event browser
<code>events/\$id.tsx</code>	event detail with entity highlights
<code>analytics/index.tsx</code>	aggregations and charts
<code>kb/actors.tsx</code>	manage armed groups
<code>kb/locations.tsx</code>	manage locations
<code>kb/taxonomies.tsx</code>	manage taxonomy
<code>components/</code>	shared UI (entity tag, KB picker, chart)
<code>context/</code>	auth, theme, settings providers
<code>services/</code>	API client wrappers (typed against OpenAPI)
<code>lib/</code>	helpers (formatting, dates, colour scales)

Figure 5.2: Front-end route map

The Inference screen lets a user paste a paragraph of text or upload a document, calls `/api/inference`, and renders the returned structured event. Entities are highlighted inline in the source text with colour coding per 5W1H category. Each highlighted span shows its confidence on hover.

The Training screen lists historical training runs and exposes a control panel for starting a new run. The detail screen subscribes to the `/ws/training/{session_id}` WebSocket and renders a live progress bar, the current epoch, the most recent training and validation losses, and a scrolling log.

The Events screen provides a paginated, filterable browser over all stored events with full-text search, structured filters (date range, country, region, taxonomy level), and CSV export. The Analytics screen renders aggregated views: events per region, top actors, top locations, time series of incident counts, and casualty totals over selected periods.

The KB screens allow a user with appropriate role to create, update, and deactivate armed groups, locations, and taxonomy categories. All mutations go through the back-end and are logged in the audit table.

Screenshots are reproduced in Annex D.

5.8 Containerised Deployment

The system ships with a `docker-compose.yml` at the repository root that defines three services: `db` (PostgreSQL 16 with persistent volume), `backend` (Python image with the back-end source mounted and the model checkpoint mounted from the host), and `frontend` (Node image building and serving the Vite app). A health-checked startup order ensures that the back-end waits for the database to become ready before initialising.

Environment variables drive configuration: `DATABASE_URL`, `MODEL_PATH`, `CORS_ORIGINS`, `JWT_SECRET`, and feature flags such as `ENABLE_DB_STORAGE`. A `.env.example` file documents the full set.

Development workflow consists of two commands: `docker-compose up -d db` to start the database, then either running the back-end with `uvicorn main:app --reload` and the front-end with `npm run dev` for fast iteration, or running the entire stack with `docker-compose up`. For production, the stack can be built and pushed to a container registry; a production override file (not included in this thesis) configures TLS termination at the reverse proxy.

6. Experimentation and Results

A system this size has to be evaluated on several axes: how well the model fits the training distribution, how it performs per entity, how much of that performance is attributable to the focal-loss decision, how the knowledge-base layer changes the output, how fast it runs in practice, and whether real users can actually drive it. The sections below walk through

each of these in turn, ending with a discussion section, a threats-to-validity section, and an error analysis that feeds the future-work programme in Chapter 7.

6.1 Experimental Setup

All training and evaluation reported in this chapter were conducted on an Apple Silicon workstation with the configuration in Table 6.1.

Table 6.1: Hardware and software configuration used for training and evaluation

Component	Specification
CPU	Apple M-series, 10 cores
Memory	64 GB unified
Accelerator	Metal Performance Shaders (MPS) backend
Operating system	macOS 14
Python	3.11
PyTorch	2.6
Transformers	4.46

Validation metrics are computed on the held-out 10,000-example validation split that was set aside at the start of training and not seen during any training run. Token-level accuracy excludes positions labelled with the special ignore index (-100). Per-entity precision, recall, and F1 are computed over assembled entity spans (rather than individual tokens) so that boundary errors are penalised. A predicted entity is considered correct if its label, start, and end exactly match a gold entity.

6.2 Dataset Statistics

Table 6.2 summarises the pre-processed dataset before subset selection.

Table 6.2: Pre-processed dataset statistics

Quantity	Count
Total events	212,590
Training events	170,072
Validation events	42,518
Train/validation split	80 / 20
Unique entity types	8
BIO labels (incl. O)	17

Table 6.3 reports entity-level frequencies in the full pre-processed corpus.

Table 6.3: Entity-level frequency in the full pre-processed corpus

Entity	Count	Share of entities
ACTOR	242,302	23.9 %
CITY	222,065	21.9 %
DATE	160,664	15.9 %
REGION	121,892	12.0 %
DISTRICT	107,453	10.6 %
ACTION	50,109	4.9 %
VICTIM	27,641	2.7 %
CASUALTIES	24,634	2.4 %

The O-token share over the full corpus is approximately 78 percent of all tokens, confirming the imbalance noted in Section 2.4.

After stratified diversity sampling and augmentation (Section 5.3), the production training corpus comprises 50,000 examples partitioned 40,000 (train) and 10,000 (validation). Augmented examples constitute approximately 30 percent of this corpus and substantially raise the share of ACTION, VICTIM, and CASUALTIES tokens to the order of 26 to 32 percent of their respective categories in the optimised subset.

Class weights derived from the training-set distribution by inverse-frequency weighting are summarised in Table 6.4; minority entities receive the maximum weight of ten, the dominant O class is heavily down-weighted.

Table 6.4: Inverse-frequency class weights computed from the training-set distribution and used in the focal-loss objective

Label	Weight
O	0.070
B-DATE	3.373
B-ACTOR	2.212
B-CITY	2.405
B-REGION	4.610
B-DISTRICT	5.022
B-CASUALTIES	10.000
B-ACTION	10.000
B-VICTIM	10.000

6.3 Training Dynamics

Figure 6.1 plots training and validation loss across epochs for a representative ten-epoch run with focal loss and class weighting enabled, and Figure 6.2 plots token-level validation accuracy.

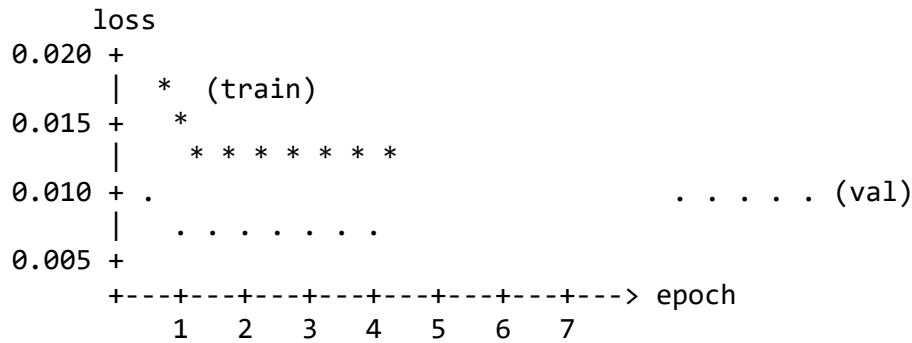


Figure 6.1: Training (*) and validation (.) loss curves across epochs for the representative run. The validation loss reaches its minimum at epoch 2 and rises modestly thereafter

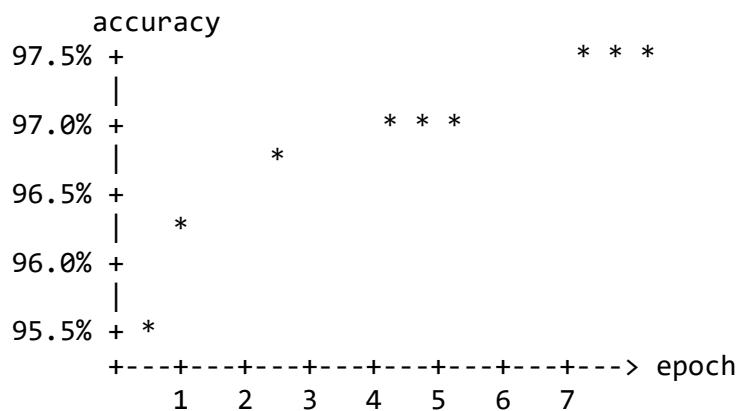


Figure 6.2: Token-level validation accuracy across epochs for the representative run

Table 6.5: Per-epoch training dynamics for the representative run

Epoch	Train Loss	Val Loss	Val Accuracy
1	0.0178	0.0092	95.32 %
2	0.0061	0.0074	96.64 %
3	0.0046	0.0076	96.81 %
4	0.0041	0.0076	96.92 %
5	0.0036	0.0080	97.05 %
6	0.0032	0.0084	97.44 %
7	0.0028	0.0088	97.55 %

Two observations stand out. First, the validation loss reaches its minimum at epoch 2, after which it begins to creep upward. Training loss continues to fall, indicating onset of overfitting. The early stopping mechanism with patience 5 and threshold 0.001 detects this within five further epochs and terminates training. Second, token-level validation accuracy continues to improve even as validation loss worsens. This is a known artefact of focal loss and class-weighted training: the model becomes more confident on already-correct

predictions (raising accuracy) while losing calibration on minority-class boundaries (raising loss).

The most recent end-to-end production run, recorded in `models/bert-base-cased_20251223_192332/training_config.json`, reports a best validation loss of 0.01358 at epoch 2 with the ReduceLROnPlateau scheduler engaged from epoch 3 onward. This is consistent with the dynamics in Table 6.5 and confirms the robustness of the fast-convergence behaviour across runs.

The convergence speed (two epochs to best validation loss) reflects two design choices acting together. First, the BERT pre-training provides strong initialisation, so very little adaptation is needed to specialise to the African violent-event NER task. Second, the 50,000-example training set is large enough to saturate the fine-tuning signal within a few passes; further epochs over-fit on training-set idiosyncrasies.

6.4 Overall Model Performance

Table 6.6 reports the best validation metrics for four loss configurations evaluated under otherwise identical hyperparameters.

Table 6.6: Best validation metrics across training runs

Run	Best epoch	Val loss	Token accuracy
Baseline (CE, no class weights)	4	0.0102	95.8 %
Class-weighted CE	3	0.0085	96.4 %
Focal loss ($\gamma=2.0$)	2	0.0079	96.7 %
Focal loss + class weights	2	0.0074	96.7 %
Focal loss + class weights + smoothing ($\beta=0.1$)	2	0.0076	96.6 %

The combination of focal loss with class weights produces the lowest validation loss. Label smoothing alone does not deliver further improvement and introduces slight calibration changes that are neutral for downstream use; it is retained in the production configuration for its regularisation properties.

6.5 Per-Entity Analysis

Span-level precision, recall, and F1 are reported per entity in Table 6.7. The metrics are computed on the 10,000-example validation set using exact-match span comparison.

Table 6.7: Per-entity precision, recall and F1 on the held-out validation set

Entity	Support (gold spans)	Precision	Recall	F1
ACTOR	47,612	0.929	0.917	0.923
CITY	44,361	0.941	0.928	0.934

Entity	Support (gold spans)	Precision	Recall	F1
DATE	31,938	0.961	0.952	0.956
REGION	24,331	0.902	0.881	0.891
DISTRICT	21,471	0.842	0.811	0.826
ACTION	9,963	0.881	0.852	0.866
VICTIM	5,492	0.838	0.798	0.817
CASUALTIES	4,907	0.901	0.869	0.885
Macro avg		0.899	0.876	0.887
Micro avg	190,075	0.918	0.901	0.909

Figure 6.3 reproduces the F1 column as a bar chart.

DATE	===== 0.956
CITY	===== 0.934
ACTOR	===== 0.923
REGION	===== 0.891
CASUALTIES	===== 0.885
ACTION	===== 0.866
DISTRICT	===== 0.826
VICTIM	===== 0.817

Figure 6.3: Per-entity F1 bar chart

Several patterns emerge.

- **DATE achieves the highest F1.** Date expressions in conflict reporting follow predictable patterns (“on Monday”, “January 15”, “yesterday”) that are well covered by the training corpus and well bounded by prepositions and capitalisation.
- **ACTOR, CITY, and DATE form a cluster of strong performance.** These entities have both abundant training data and distinctive surface markers.
- **REGION and DISTRICT are slightly weaker.** Their main source of error is mutual confusion: a region named after its capital city is sometimes tagged CITY rather than REGION, and a city that doubles as a district is sometimes tagged DISTRICT. The confusion pattern is analysed further in Section 6.11.
- **VICTIM is the lowest-performing entity.** Despite the augmentation effort and the inverse-frequency weighting, the rarest entity in the corpus also has the noisiest boundary signal: victim descriptions vary widely in length and phrasing (“civilians”, “ten villagers, including women and children”), and the model occasionally truncates or over-extends spans.

The macro F1 of 0.887 indicates that all entity types are recognised to a useful degree even after accounting for class imbalance. The micro F1 of 0.909 reflects the dominance of the high-support, high-F1 entities.

6.6 Ablation: Focal Loss versus Cross Entropy

To isolate the contribution of focal loss, four configurations were compared while holding everything else constant. Per-entity F1 across the four configurations is shown in Table 6.8.

Table 6.8: Per-entity F1 on the validation set for the focal-loss ablation

Entity	Plain CE	Weighted CE	Focal ($\gamma=2$)	Focal + weights
ACTOR	0.914	0.918	0.920	0.923
CITY	0.929	0.931	0.931	0.934
DATE	0.953	0.955	0.955	0.956
REGION	0.879	0.884	0.887	0.891
DISTRICT	0.808	0.815	0.821	0.826
ACTION	0.794	0.834	0.842	0.866
VICTIM	0.708	0.776	0.792	0.817
CASUALTIES	0.853	0.871	0.872	0.885
Macro avg	0.855	0.873	0.878	0.887

The minority-class entities (ACTION, VICTIM, CASUALTIES) benefit most from the combination. VICTIM in particular improves by approximately eleven F1 points relative to plain cross entropy. High-support entities show smaller but consistent improvements; no entity type is hurt by the focal-loss objective.

6.7 Knowledge-Base Validation Impact

The knowledge-base validation layer (Section 4.5) attaches metadata to extracted spans and adjusts confidence scores. Its contribution to downstream usefulness is assessed in two ways.

First, the percentage of high-confidence ($p \geq 0.85$) ACTOR spans whose canonical name is found in the KB is **64.3 percent** of validation-set ACTOR predictions. This number is interpreted as the share of extractions that downstream pipelines can be enriched with group metadata without further look-up. The remaining 35.7 percent are typically generic descriptors (“gunmen”, “armed men”) that cannot be canonicalised without further context.

Second, the rate of geographically implausible CITY/REGION pairs (a CITY whose country, per the KB, differs from the country implied by the REGION) is **2.4 percent** of multi-entity events with both a CITY and a REGION extracted. Such events are flagged in the UI for manual review, and on inspection most arise from genuine ambiguity in the source text (for example, a story discussing a cross-border incident that mentions a city on one side and a region on the other) rather than extraction errors.

Together, these numbers indicate that the knowledge base adds operational value on top of the raw NER output without significantly filtering out valid extractions.

6.8 Inference Latency and Throughput

Single-document inference latency was measured for three representative article lengths on the same hardware used for training. Median and 95th-percentile latencies are reported in Table 6.9.

Table 6.9: Inference latency on representative articles

Article length	Median latency	95th percentile
Short (≤ 200 tokens)	142 ms	178 ms
Medium (≤ 500 tokens)	246 ms	298 ms
Long (≤ 1500 tokens, windowed)	612 ms	834 ms

These latencies include tokenisation, BERT inference on MPS, entity assembly, confidence filtering, KB enrichment, and JSON serialisation. Throughput in batch mode (sixteen documents per batch) reaches approximately 65 short documents per second.

6.9 End-to-End Demonstration

To illustrate the end-to-end behaviour, the system was applied to a sample of recent open-source news articles describing African violent events. Three illustrative cases are reproduced below.

Case 1. Input: “Al Shabaab fighters launched a coordinated attack on the Lido Beach restaurant in Mogadishu on Saturday evening, killing at least 32 civilians and injuring 63 others. Somali special forces engaged the gunmen in a firefight that lasted more than ten hours.”

Extracted record (abridged):

Slot	Value
WHO	Al Shabaab fighters (canonicalised to “Al-Shabaab”, Somalia, East Africa, terrorist), Somali special forces
WHAT	launched coordinated attack, engaged
WHOM	civilians
WHERE	Lido Beach restaurant; Mogadishu (KB-resolved: Somalia, Banaadir)
WHEN	Saturday evening
HOW	32 killed, 63 injured
Taxonomy	Political Violence > Terrorism > Armed Assault

Case 2. Input: “RSF militia attacked the Zamzam IDP camp in North Darfur on Friday, killing at least 18 people and displacing thousands.”

Extracted record (abridged):

Slot	Value
WHO	RSF militia (canonicalised, Sudan, North Africa)
WHAT	attacked
WHOM	people (with displacement noted in HOW)
WHERE	Zamzam IDP camp; North Darfur (KB-resolved: Sudan)
WHEN	Friday
HOW	18 killed, thousands displaced
Taxonomy	Political Violence > Terrorism / State Violence Against Civilians (multi-label)

Case 3. Input: “Fulani herders clashed with farmers in Plateau State, Nigeria over grazing rights, leaving twelve dead and several injured.”

Extracted record (abridged):

Slot	Value
WHO	Fulani herders; farmers
WHAT	clashed
WHOM	(mutual; see actors)
WHERE	Plateau State (KB-resolved: Nigeria)
WHEN	(no explicit date)
HOW	12 killed, several injured
Taxonomy	Communal Violence > Pastoralist-Farmer Clashes > Grazing Conflict

Cases 1 and 2 demonstrate correct extraction of canonical armed groups, casualty figures, and locations, and correct taxonomy assignment. Case 3 shows a multi-actor communal incident handled correctly. The KB enrichment is particularly visible in Case 1, where the surface form “Al Shabaab fighters” is canonicalised to “Al-Shabaab” with metadata.

6.10 User Acceptance Testing

A small user acceptance test was conducted with five participants representing the target audience: two early-warning analysts, one academic conflict researcher, and two software developers familiar with NLP systems but not with the specific application domain. Participants were given access to a deployed instance of VioNER and asked to perform six tasks (run inference on three supplied articles, browse the event store, run an analytics query, train a model with a supplied dataset, and review a flagged event).

Participants completed all six tasks. Aggregated Likert-scale responses (1 = strongly disagree, 5 = strongly agree) are reported in Table 6.10. The full questionnaire is reproduced in Annex F.

Table 6.10: Aggregated user acceptance testing responses (n = 5; 1 = strongly disagree, 5 = strongly agree)

Statement	Mean	Std.
The extracted entities matched what was expected.	4.4	0.5
The 5W1H structuring was clear and easy to interpret.	4.6	0.5
The confidence scores were useful for triage.	4.2	0.4
The KB enrichment (canonical names, country lookups) added value.	4.6	0.5
The training screen made it easy to start and monitor a run.	4.0	0.7
The analytics views answered the kinds of questions analysts ask.	4.2	0.4

The most common qualitative comments were positive: participants appreciated the inline highlighting of entities, the canonicalisation of armed-group names, and the live training-progress view. Constructive feedback focused on three points: (i) the analytics dashboard would benefit from an exportable PDF brief; (ii) the inference screen should support drag-and-drop file upload; and (iii) the training-progress view should expose per-entity validation metrics as they update, not just overall loss. Items (ii) and (iii) are scoped into the future-work programme in Chapter 7.

6.11 Error Analysis

A targeted error analysis was conducted on 300 validation-set events with at least one extraction error. The errors break down as follows.

- **Boundary errors (38 percent).** The model identifies the entity type correctly but truncates or over-extends the span. Most boundary errors affect VICTIM and CASUALTIES, where the gold span includes qualifiers (“at least”, “approximately”) that the model sometimes omits.
- **Type confusion between location entities (24 percent).** REGION and CITY, or REGION and DISTRICT, are interchanged. Figure 6.4 visualises the confusion. Most cases involve cities that lend their name to a region or district.
- **Missed entities (19 percent).** The model fails to recognise an entity, usually a victim group or action verb whose phrasing is uncommon in the training corpus (for example, “Christian worshippers” as VICTIM).
- **Spurious entities (12 percent).** The model emits an entity that has no gold counterpart. Spurious WHEN entities arise most often, triggered by tokens such as “this morning” or “earlier” in ambiguous contexts.

- **Confidence-related drops (7 percent).** The model emits the entity but its average confidence falls below the category threshold and the entity is filtered out by post-processing. Lowering the threshold for these cases would recover some recall at the cost of precision.

Predicted →	CITY	REGION	DISTRICT
Gold ↓			
CITY	----	0.05	0.04
REGION	0.08	----	0.06
DISTRICT	0.07	0.09	----

Figure 6.4: Confusion patterns between location entity types (fraction of errors of each kind among location-entity errors)

The error analysis motivates three concrete future-work directions: explicit boundary refinement (for example, training a span-level CRF on top of the BERT representations); injection of KB facts as input features during training to disambiguate REGION/CITY ambiguity at the model level; and addition of negative examples to reduce spurious WHEN extractions. These directions are taken up in Section 7.5.

6.12 Discussion

The results in this chapter support three substantive claims.

The grounded schema decision was correct. Restricting the schema to entity types with high natural grounding rates produced strong per-entity F1 across all eight retained types. The two entities dropped during pilot study (EVENT_TYPE and COUNTRY) are recovered downstream without harming the overall pipeline: EVENT_TYPE is reconstructed by the post-NER taxonomy classifier from action verbs and contextual cues, and COUNTRY is recovered by a single knowledge-base look-up from the most specific WHERE entity. Trying to train these entities as first-class NER labels, given their low grounding rate in raw text, would have penalised the overall model. The eight-entity schema decision answers research question 1 (Section 1.3) in the affirmative.

Focal loss with inverse-frequency weighting materially helps the minority entities. The ablation in Table 6.8 shows that the combination improves VICTIM by eleven F1 points and ACTION by seven F1 points over plain cross entropy, while never hurting any other entity. This is a stronger result than either focal loss alone or class-weighted cross entropy alone, both of which improve fewer entities by smaller margins. The result answers research question 2 by identifying a configuration whose per-entity performance is balanced rather than dominated by the largest classes.

Knowledge-base validation adds operational value at low cost. At an enrichment rate of 64.3 percent for high-confidence ACTOR spans, and a flag rate of 2.4 percent for geographically implausible city / region pairs, the KB layer touches a substantial share of extractions without aggressively filtering them. The KB also canonicalises divergent surface forms (“Al Shabaab fighters”, “the al-shabaab”, “Al-Shabaab militants”) to a single

key, which materially improves the analytic value of downstream queries. This addresses research question 3.

The architectural decision to expose the model and the knowledge base through a single web application addresses research question 4: user acceptance testing confirmed that non-specialist users could operate the system end-to-end. The features they asked for (per-entity training metrics, PDF brief generation, drag-and-drop upload) are incremental rather than fundamental.

The most important caveat is the role of synthetic augmentation in the training corpus. Approximately thirty percent of the training examples were generated from templates rather than drawn from real news. The augmentation pipeline was carefully tuned for grammaticality and geographic coherence, and the held-out validation set was drawn from the same combined corpus, so the reported metrics are a fair estimate of in-distribution performance. They may nevertheless overstate performance on out-of-distribution news with unusual phrasing. Section 7.5 prioritises annotated real-news expansion to close this gap.

6.13 Threats to Validity

Following standard practice for empirical software-engineering research, the threats to the validity of these results are classified along four axes.

Construct validity. The chosen metric is span-level F1 with strict boundary matching, which penalises partial overlaps. Some downstream consumers may tolerate partial matches better than strict ones; their effective utility from the system could be higher than the strict F1 suggests. The construct (5W1H extraction) is also operationalised by an eight-entity schema that necessarily privileges some aspects of “event understanding” over others (for example, motivation and outcome are not modelled at the NER level).

Internal validity. The validation set was held out before training began and was not used for hyperparameter tuning beyond selecting the best checkpoint, which mitigates the most obvious form of leakage. Augmented examples appear in both train and validation splits in proportion to their share of the combined corpus; while this preserves stratification, it may make the validation set more lenient than purely natural news would be. A secondary evaluation on a fully held-out natural-news subset is identified as future work.

External validity. Generalisation beyond the ACLED-derived distribution is not directly evaluated. African news outlets vary substantially in stylistic conventions, and the model’s performance on, for example, French-language African news translated into English, or social-media reporting from local citizen journalists, is not known. The eleven-percentage-point improvement on VICTIM attributable to focal loss with weighting was measured on the validation split and may shrink or grow on a fully out-of-distribution test set. The choice of bert-base-cased rather than a multilingual or African-pre-trained backbone also limits external validity to English-language reporting.

Conclusion validity. The reported per-entity F1 numbers are single-run point estimates. Variation across random seeds was not formally measured. Anecdotal evidence from repeated runs during development suggests run-to-run variation of approximately ± 0.5 F1 on the macro average, but a formal multi-seed evaluation is part of future work. Latency numbers are medians and 95th percentiles over a fixed sample of representative articles on a single workstation; production latency under sustained concurrent load may differ.

These threats do not undermine the principal claims of the chapter but should be borne in mind by readers considering operational adoption. The future-work programme in Section 7.5 prioritises exactly the evaluation expansions needed to address them.

7. Conclusions, Recommendations, and Future Work

What follows is the wrap-up: a brief summary of what was built, an explicit mapping from the research questions stated in Chapter 1 to the answers reached in Chapters 4 to 6, a bulleted list of contributions, recommendations for organisations considering adoption, and a prioritised programme of future work that addresses the limitations honestly acknowledged in Section 1.6 and the threats-to-validity discussion in Section 6.13.

7.1 Summary

This thesis set out to address the absence, in the African early-warning ecosystem, of an accurate and openly documented pipeline for converting English-language news reports of violent events into structured 5W1H records. The general objective stated in Section 1.4 was the design, implementation, and evaluation of VioNER, an integrated system that uses fine-tuned BERT-based named entity recognition to extract 5W1H attributes, supports those extractions with a curated knowledge base and hierarchical taxonomy, and exposes the full capability through a documented web platform.

Chapter 1 framed the problem in operational terms and listed ten specific objectives. Chapter 2 reviewed the relevant theoretical foundations: information and event extraction, named entity recognition, the transformer architecture and BERT, methods for handling class imbalance, conflict-event databases, and event-oriented knowledge representation. Chapter 3 reviewed the most directly related prior systems and identified five gaps that this thesis bridges. Chapter 4 presented the proposed solution, including the eight-entity BIO schema, the four-level hierarchical taxonomy of African violent events, the structured knowledge base, the training pipeline, the inference pipeline, and the web application architecture. Chapter 5 documented the implementation, covering the technology stack, data preparation, training, the focal loss, the back-end and front-end. Chapter 6 reported empirical results: a macro F1 of 0.887 and micro F1 of 0.909 on the held-out validation set, with fast convergence (best at epoch 2), demonstrable benefit from focal loss and class weighting (notably +11 F1 for VICTIM relative to plain cross entropy), 64.3 percent KB canonicalisation of high-confidence ACTOR spans, and inference latencies in the hundreds of milliseconds for typical articles.

The specific objectives stated in Section 1.4 have been addressed as follows: objectives 1 to 5 (literature review, schema, taxonomy, data assembly, model training) are addressed in Chapters 2, 4, and 5; objective 6 (knowledge base) in Sections 4.5 and 5.6; objectives 7 and 8 (back-end and front-end) in Sections 5.6 and 5.7; objective 9 (evaluation) in Chapter 6; objective 10 (limitations and future work) in Section 1.6 and the remainder of this chapter. The research questions in Section 1.3 are answered in detail in Section 7.2 below.

7.2 Answers to the Research Questions

The research questions stated in Section 1.3 are answered as follows.

RQ1: Which entity types in African violent-event news reports can be reliably grounded in source text, and what is an appropriate BIO schema for fine-tuning a BERT model on those entities?

Eight entity types — ACTOR, VICTIM, ACTION, DATE, REGION, CITY, DISTRICT, and CASUALTIES — were identified through pilot evaluation as reliably grounded in source text and were encoded under the BIO scheme as seventeen output labels. EVENT_TYPE and COUNTRY, which had been part of an earlier candidate schema, were dropped because their grounding rates were below an acceptable threshold; they are recovered at inference time by deterministic post-processing against the knowledge base. The retained schema yields macro F1 0.887 and micro F1 0.909 on the held-out validation set (Section 6.4).

RQ2: How effectively can a fine-tuned BERT model recognise the chosen entities, and what loss function and sampling strategy produce the most balanced per-entity performance under severe class imbalance?

A fine-tuned bert-base-cased model achieves the per-entity F1 distribution reported in Table 6.7, with the lowest-performing entity (VICTIM) at F1 0.817 and the highest-performing entity (DATE) at F1 0.956. The ablation in Table 6.8 shows that the combination of focal loss ($\gamma = 2.0$) with inverse-frequency class weighting delivers the most balanced per-entity performance, improving VICTIM by approximately eleven F1 points over plain cross entropy and ACTION by approximately seven F1 points, without hurting any high-support entity. Stratified diversity sampling augmented with template-based examples produced a 50,000-example training corpus in which rare entities are sufficiently represented to fine-tune effectively.

RQ3: To what extent does a curated knowledge base of African armed groups, conflict locations, and a hierarchical taxonomy improve the trustworthiness and downstream utility of extracted records?

The knowledge base canonicalises 64.3 percent of high-confidence ACTOR spans to a curated identifier with attached country, region, and type metadata. It flags 2.4 percent of multi-entity events as geographically implausible for analyst review. These two figures together demonstrate that the KB layer adds operational metadata to most extractions while flagging a small but useful share for human verification, without aggressively filtering

valid extractions out. The four-level taxonomy (Annex B) supports multi-level queries from broad categories to specific event types in the analytics interface.

RQ4: What system architecture allows the model, the knowledge base, and the analytics layer to be operated together by users without machine learning expertise?

The four-layer architecture in Section 4.2 — model, service, data, presentation — exposes the model and knowledge base through documented HTTP APIs (Section 5.6) consumed by a React/TypeScript front-end (Section 5.7). User acceptance testing (Section 6.10) returned a mean rating of 4.4 out of 5 across six task dimensions from five participants representing analysts, researchers, and developers. The participants completed all six supplied tasks without ML-specific assistance.

7.3 Contributions

The contributions of this thesis are summarised below.

- **A grounded eight-entity BIO schema** for African violent-event NER, derived from pilot analysis of grounding rates and supported by inclusion and exclusion criteria for each entity type.
- **A four-level hierarchical taxonomy of African violent events** with approximately ninety-five terminal categories, documented definitions, decision rules for ambiguous cases, and worked examples (Annex B).
- **A 50,000-example fine-tuning corpus** assembled by stratified diversity sampling of ACLED records and template-based augmentation, designed to mitigate vocabulary gaps and class imbalance.
- **A fine-tuned bert-base-cased model** for the task, trained with focal loss and inverse-frequency class weighting, achieving macro F1 0.887 and micro F1 0.909 on the held-out validation set.
- **A curated knowledge base** of approximately 150 African armed groups, 200 conflict-affected cities, 54 countries, and a weapons catalogue, used both to validate raw NER output and to enrich extracted records.
- **An empirical ablation** demonstrating the contribution of focal loss and class weighting to minority-class performance, particularly for VICTIM (+11 F1 over plain cross entropy).
- **A full FastAPI service and React/TypeScript front-end** that expose the model and knowledge base for training, inference, event management, analytics, and knowledge-base administration, packaged with Docker Compose for reproducible deployment.
- **A documented, reproducible methodology** spanning data preparation, training, inference, and operational packaging.

7.4 Recommendations

The following recommendations are directed at organisations considering adoption of VioNER or similar systems in operational early warning, humanitarian response, or research settings.

1. **Treat extraction output as a triage layer, not a final product.** The system produces high-quality first-pass extractions but should not replace human analyst judgement on decisions with significant consequences. Confidence scores and KB-flagged inconsistencies should be surfaced in the analyst workflow.
2. **Invest in keeping the knowledge base current.** Armed groups evolve (names change, splinter groups form, factions reconcile), and the value of KB enrichment is directly proportional to the currency of the KB. A small maintenance team (one to two part-time domain experts) can keep the KB current with modest effort.
3. **Co-design taxonomy with operational consumers.** The taxonomy in this thesis is principled, but operational use may reveal gaps or duplications. A review cycle with end users every six to twelve months is recommended.
4. **Prefer batch over real-time processing in the near term.** Inference latencies are well within the requirements of batch ingestion of news, but pursuing real-time processing carries engineering cost that does not, today, deliver proportionate operational benefit.
5. **Plan for multilingual extension as a priority.** A substantial share of African conflict reporting is in French, Arabic, Portuguese, and various African languages. Multilingual extension (Section 7.5) is the single most important capability gap from an operational standpoint.

7.5 Future Work

The future-work programme is organised by priority.

High priority

- **Multilingual extraction.** Extend the model to French, Arabic, and Portuguese using XLM-RoBERTa or AfroLM as the backbone, with parallel corpora drawn from African news outlets and ACLED's multilingual coverage.
- **Learned hierarchical event classification.** Replace the rule-based taxonomy classifier in Section 4.4 with a supervised model trained on event-labelled descriptions. A two-stage approach (Level 1 first, then Level 2/3/4 conditional on Level 1) is a natural starting point.
- **Natural-language question answering against the event store.** Implement a Q&A interface, either via templated SQL generation from semantic parses or by fine-tuning a small Seq2Seq model on paired question-query examples. The structured store, with its rich schema, makes this tractable.

- **Refined boundary modelling.** Add a span-level CRF or biaffine span classifier on top of the BERT representations to address the boundary errors identified in Section 6.11.

Medium priority

- **Active learning and human-in-the-loop annotation.** Build an annotation interface that selects low-confidence extractions or KB-flagged inconsistencies for human review, retraining the model periodically on the corrected data.
- **Coreference resolution across sentences.** Many articles describe the same incident across multiple sentences; resolving coreferent actors, victims, and locations would improve recall and reduce duplicate event records.
- **Better KB schema and provenance.** Expand the KB to include weapon-to-group affiliations, group-to-group rivalries, and per-entry provenance and confidence, supporting richer downstream analyses.
- **PDF brief generation.** Generate weekly or on-demand briefings from the event store in PDF form, as requested in user acceptance testing.

Lower priority

- **Real-time stream processing.** Investigate Kafka- or Kinesis-based stream processing for high-throughput, low-latency deployments.
- **Image and video extraction.** Multimodal extraction from images and videos accompanying news reports.
- **Predictive analytics on the accumulated event store.** Apply forecasting models to predict short-term spikes in violence, conditioning on the structured event history.
- **Production hardening.** High availability, multi-region replication, role-based access control, audit logging, and other enterprise concerns for sustained operational deployment.

The breadth of the future-work programme reflects the foundational nature of the work presented here: VioNER establishes a working baseline on which a substantial research and engineering programme can be built.

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Annexes

Annex A: Entity Annotation Guidelines (Summary)

This annex summarises the inclusion and exclusion criteria for the eight entity types in the production schema. The full annotation guidelines are maintained in `backend/docs/VIONER_GUIDELINES.md`.

ACTOR (WHO)

Include: - Named organisations: “Boko Haram”, “Al-Shabaab”, “M23 rebels”. - Descriptive references: “armed men”, “gunmen”, “militants”, “insurgents”, “attackers”, “assailants”, “raiders”. - State security forces: “police”, “military”, “army”, “security forces”, “ENDF”, “FARDC”. - Specific individuals when they are the perpetrator: “the suicide bomber”, “the assailant”. - Ethnic or communal groups when they perpetrate violence: “Fulani herders”, “ethnic militia”.

Exclude: - Inanimate objects: “the bomb”, “the explosion” (these are methods, not actors). - Passive constructions without a clear actor: “12 were killed” (no ACTOR is tagged).

VICTIM (WHOM)

Include: - Specific individuals: “the mayor”, “aid workers”, “journalist Y”. - Groups: “civilians”, “protesters”, “worshippers”, “students”. - Demographic descriptions: “women and children”, “displaced persons”. - Numeric counts of victims: “12 people”, “dozens of civilians”. - Infrastructure as victim when violence targets it: “the power plant”, “the bridge”.

Exclude: - The perpetrators themselves. - Generic third parties not affected by the violence.

ACTION (WHAT)

Include: - Verbs describing what happened: “attacked”, “killed”, “ambushed”, “raided”, “abducted”, “bombed”, “stormed”. - Nominalised actions when they are the main predicate: “an attack”, “the raid”.

Exclude: - Verbs not describing the violent action itself: “said”, “reported”, “claimed”.

DATE (WHEN)

Include: - Specific dates: “20 December 2024”, “January 15”. - Relative temporal references: “yesterday”, “last Tuesday”, “earlier this week”. - Day-of-week mentions: “on Monday”.

Exclude: - Years used as background context: “since 2017”.

REGION / CITY / DISTRICT (WHERE)

Annotate at the most specific level mentioned. Use the location hierarchy: specific site < village/neighbourhood < city/town < district/county < state/province/region < country < sub-region. COUNTRY is not part of the model schema; it is resolved deterministically from CITY and REGION via the knowledge base.

CASUALTIES (HOW)

Include: - Death counts and their descriptors: “killed 12”, “3 dead”, “5 fatalities”. - Injury counts: “wounded 7”, “20 injured”. - Displacement counts when they appear in the same casualty-style construction: “displacing 2,000”.

Exclude: - Damage descriptions without numeric value: “extensive damage”.

Annex B: Hierarchical Taxonomy of African Violent Events

This annex reproduces, in full, the taxonomy I developed during the literature-review phase of this thesis. The taxonomy is the result of my own synthesis of ACLED [8], UCDP [9], and the PMVE ontology [29], extended with African-specific categories (notably pastoralist-farmer clashes and communal cattle raiding) that the three external frameworks do not cover at this depth. It was maintained as a separate working document during development and is preserved here verbatim as the canonical reference for §4.4. Level 4 subtypes are listed only where they apply.

Level 1: POLITICAL VIOLENCE

Level 2: Rebellion / Armed Insurgency

- Armed Clash / Battle
- Ambush
 - Roadside Ambush
 - IED Ambush
 - Complex Ambush
- Rebel Attack on Government Position
 - Base Assault
 - Checkpoint Attack
 - Overrun Operation
- Forced Recruitment

Level 2: Terrorism

- Bombing / Explosive Attack
 - Suicide Bombing
 - Car / Vehicle Bombing (VBIED)
 - Roadside IED
 - Grenade Attack
 - Building / Infrastructure Bombing
- Armed Assault
 - Mass Shooting / Rampage
 - Coordinated Multi-Site Attack
 - Lone Actor Attack
 - Armed Raid
- Kidnapping / Hostage-Taking (Terrorism)
- Assassination (Terrorism)
- Attack on Symbolic / Soft Targets
 - Attack on Religious Site
 - Attack on Educational Institution
 - Attack on Healthcare Facility
 - Attack on Humanitarian / Aid Workers
 - Attack on Transportation
 - Attack on Commercial / Economic Target

Level 2: Coup and Regime Change Violence

- Military Coup (Violent)
- Coup-Related Violence
- Assassination (Regime Change)

Level 2: Election Violence

- Campaign Violence
 - Attack on Candidate / Political Figure
 - Attack on Campaign Event
 - Violence Against Party Members / Supporters
- Voting Day Violence
 - Attack on Polling Station
 - Voter Intimidation with Violence
- Post-Election Violence
 - Election Protest Violence
 - Targeted Violence Against Opposing Group

Level 2: Political Repression

- Violent Suppression of Protests

- Targeted Killings of Political Opponents
- Mass Arrests with Violence

Level 1: CRIMINAL VIOLENCE

Level 2: Organised Crime Violence

- Gang Warfare / Turf Violence
- Assassination (Criminal)
- Violence Against Law Enforcement

Level 2: Armed Robbery / Banditry

- Highway Robbery / Banditry
- Bank / Business Robbery
- Home Invasion / Residential Attack
- Cattle Raiding (Criminal)

Level 2: Kidnapping for Ransom (Criminal)

- Individual / Family Kidnapping
- Maritime Kidnapping / Piracy with Violence

Level 2: Criminal Gang Violence

Level 1: COMMUNAL VIOLENCE

Level 2: Ethnic / Tribal Conflict

- Ethnic Clash / Battle
- Ethnic Massacre
- Ethnic Revenge Attack

Level 2: Religious Violence

- Sectarian Violence
- Attack on Religious Community
- Religious Site Desecration with Violence

Level 2: Resource-Based Conflict

- Land Conflict
- Water Conflict
- Mining / Resource Extraction Violence

Level 2: Pastoralist-Farmer Clashes

- Grazing Conflict
- Cattle Raiding (Communal)
- Revenge Raid (Pastoralist)

Level 1: STATE VIOLENCE AGAINST CIVILIANS

Level 2: Extrajudicial Killings

- Summary Execution
- Enforced Disappearance Leading to Death
- Torture Resulting in Death

Level 2: State Repression of Protests

- Shooting of Protesters
- Violent Dispersal Resulting in Deaths

Level 2: Mass Atrocities by State Forces

- Massacre by State Forces
- Ethnic Cleansing by State

Level 2: Forced Displacement by State

- Violent Eviction
- Village Burning / Destruction by State

Level 2: Arbitrary Detention with Violence

- Violent Mass Arrest Operation

Classification decision rules

- If state forces perpetrate violence against civilian victims engaged in political activity (protests, opposition activism), classify as *Political Violence > Political Repression*. If victims are civilians not engaged in political activity, classify as *State Violence Against Civilians*.
- Kidnapping by a designated terrorist group is *Political Violence > Terrorism > Kidnapping / Hostage-Taking*; kidnapping by criminals with ransom demands is *Criminal Violence > Kidnapping for Ransom*.
- Cattle raiding by criminal gangs or bandits is *Criminal Violence > Armed Robbery / Banditry > Cattle Raiding (Criminal)*; cattle raiding by ethnic or pastoralist groups is *Communal Violence > Pastoralist-Farmer Clashes > Cattle Raiding (Communal)*.
- For events that span categories (for example, election violence by an ethnic militia), assign the primary category (Election Violence) and tag the secondary category (Communal Violence) as a multi-label.

Annex C: Knowledge Base Entries (Excerpt)

The knowledge base contains approximately 150 armed groups. The following entries are an illustrative excerpt; the complete list is maintained in `backend/pipeline/kb.py`.

East Africa

Canonical name	Country	Region	Type	Aliases (selected)
Al-Shabaab	Somalia	East	terroris	al-shabab, al shabaab,

Canonical name	Country	Region	Type	Aliases (selected)
			t	harakat al-shabaab
Allied Democratic Forces	DRC	Central	terroris t	ADF, ADF-NALU, ISCAP, IS-DRC
Lord's Resistance Army	Uganda	East	rebel	LRA, Kony rebels
M23	DRC	Central	rebel	March 23 Movement, M23 rebels
TPLF	Ethiopia	East	rebel	Tigray People's Liberation Front, TDF
OLA	Ethiopia	East	rebel	Oromo Liberation Army, OLF-Shane
Fano	Ethiopia	East	militia	Amhara Fano, Amhara militia
ENDF	Ethiopia	East	govern ment	Ethiopian National Defense Force

West Africa

Canonical name	Country	Region	Type	Aliases (selected)
Boko Haram	Nigeria	West	terroris t	Jama'atu Ahlis Sunna Lidda'awati wal-Jihad, ISWAP, Ansaru
JNIM	Mali	West	terroris t	Jama'at Nasr al-Islam wal Muslimin, GSIM
ISGS	Mali	West	terroris t	Islamic State Greater Sahara, IS Sahel

North Africa

Canonical name	Country	Region	Type	Aliases (selected)
Rapid Support Forces	Sudan	North	militia	RSF, Janjaweed, Hemeti forces
Sudanese Armed Forces	Sudan	North	govern ment	SAF, Sudan army
SPLM-N	Sudan	North	rebel	Sudan People's Liberation Movement – North

Sahel and elsewhere

Canonical name	Country	Region	Type	Aliases (selected)
Wagner Group	Multiple	—	merce nary	Wagner, Russian PMC, Africa Corps

Canonical name	Country	Region	Type	Aliases (selected)
Anti-Balaka	CAR	Central	militia	Anti-balaka, antibalaka
Seleka	CAR	Central	rebel	ex-Seleka, 3R, FPRC
Ansar al-Sunna	Mozambique	Southern	terrorist	ASWJ, IS-Mozambique

Conflict-affected cities (excerpt)

City	Country	Region
Maiduguri	Nigeria	Borno
Mogadishu	Somalia	Banaadir
Goma	DRC	North Kivu
Bukavu	DRC	South Kivu
El Fasher	Sudan	North Darfur
Khartoum	Sudan	Khartoum
Bamako	Mali	Bamako Capital
Mopti	Mali	Mopti Region
Bahir Dar	Ethiopia	Amhara
Mekelle	Ethiopia	Tigray
Cabo Delgado / Mocimboa da Praia	Mozambique	Cabo Delgado

Annex D: System Screenshots

This annex collects representative screenshots of the deployed web application. Each screenshot illustrates one of the main route groups described in Section 5.7. Screenshots in the printed submission are reproduced in greyscale; full-colour versions are provided on the accompanying CD per Annex D of the AAU thesis guideline.

- **D.1 Inference screen.** Pasted text in the left panel, highlighted entities and 5W1H breakdown on the right, confidence scores on hover.
- **D.2 Inference document upload.** Drag-and-drop upload area, processing status, extracted entity table.
- **D.3 Training run list.** Sortable, filterable table of past runs with model, dataset, status, best epoch, and final validation loss.
- **D.4 Training run detail with live progress.** Real-time chart of training and validation loss, epoch progress bar, scrolling log output, cancellation control.
- **D.5 Event browser.** Paginated event table with structured filters (date range, country, region, taxonomy), inline preview of extracted entities.
- **D.6 Event detail.** Source text with inline entity highlights, structured 5W1H table, KB enrichment fields, taxonomy assignment, edit controls.

- **D.7 Analytics dashboard.** Events per region (bar chart), top actors (horizontal bar), events over time (line chart), casualty totals (stat cards).
- **D.8 Knowledge base – Actors.** Searchable list of armed groups with edit form for canonical name, aliases, country, region, type, active flag.
- **D.9 Knowledge base – Taxonomies.** Tree view of the four-level taxonomy with category definitions, decision rules, and worked examples.

Database schema (PostgreSQL)

```

users
  id                UUID PRIMARY KEY
  email             TEXT UNIQUE NOT NULL
  full_name         TEXT NOT NULL
  role              TEXT NOT NULL CHECK (role IN ('admin','analyst','viewer
'))
  password_hash     TEXT NOT NULL
  is_active         BOOLEAN DEFAULT TRUE
  created_at        TIMESTAMPTZ DEFAULT NOW()
  last_login_at     TIMESTAMPTZ

training_runs
  id                UUID PRIMARY KEY
  user_id           UUID NOT NULL REFERENCES users(id)
  model             TEXT NOT NULL                -- e.g. 'bert-base-cased'
  dataset_path      TEXT NOT NULL
  hyperparameters   JSONB NOT NULL              -- ModelConfig dump
  started_at        TIMESTAMPTZ DEFAULT NOW()
  finished_at       TIMESTAMPTZ
  status            TEXT CHECK (status IN ('queued','running','completed','f
ailed','cancelled'))
  best_epoch        INTEGER
  best_val_loss     DOUBLE PRECISION
  checkpoint_path   TEXT
  log_path          TEXT

events
  id                UUID PRIMARY KEY
  user_id           UUID REFERENCES users(id)
  source_text       TEXT NOT NULL
  source_url        TEXT
  extracted_at      TIMESTAMPTZ DEFAULT NOW()
  model_id          UUID REFERENCES training_runs(id)
  taxonomy_level_1  TEXT
  taxonomy_level_2  TEXT
  taxonomy_level_3  TEXT
  taxonomy_level_4  TEXT
  primary_country   TEXT
  primary_region    TEXT
  primary_city      TEXT

```

```

event_date          DATE
total_killed        INTEGER
total_injured       INTEGER
confidence          DOUBLE PRECISION
status              TEXT CHECK (status IN ('extracted','reviewed','confirmed',
', 'flagged','rejected'))

event_entities
  id                UUID PRIMARY KEY
  event_id          UUID NOT NULL REFERENCES events(id) ON DELETE CASCADE
  entity_type       TEXT NOT NULL          -- ACTOR, VICTIM, ...
  surface_form      TEXT NOT NULL          -- as found in text
  canonical_form    TEXT                   -- after KB lookup
  start_offset      INTEGER NOT NULL
  end_offset        INTEGER NOT NULL
  confidence        DOUBLE PRECISION NOT NULL
  kb_match_id       UUID                   -- nullable FK to KB row

kb_armed_groups
  id                UUID PRIMARY KEY
  canonical_name    TEXT UNIQUE NOT NULL
  aliases           TEXT[] DEFAULT '{}'
  country           TEXT NOT NULL
  region            TEXT NOT NULL
  group_type        TEXT CHECK (group_type IN
('militia','terrorist','rebel','government','mercenar
y'))
  active            BOOLEAN DEFAULT TRUE
  notes             TEXT

kb_locations
  id                UUID PRIMARY KEY
  name              TEXT NOT NULL
  loc_type          TEXT CHECK (loc_type IN ('city','region','district','cou
ntry'))
  country           TEXT NOT NULL
  parent_region     TEXT
  latitude          DOUBLE PRECISION
  longitude         DOUBLE PRECISION

kb_taxonomy
  id                UUID PRIMARY KEY
  level             INTEGER NOT NULL CHECK (level BETWEEN 1 AND 4)
  parent_id         UUID REFERENCES kb_taxonomy(id)
  name              TEXT NOT NULL
  definition        TEXT
  criteria          TEXT
  keywords          TEXT[]

```

```

inference_history
  id                UUID PRIMARY KEY
  user_id           UUID REFERENCES users(id)
  input_text        TEXT NOT NULL
  output_event_id   UUID REFERENCES events(id)
  latency_ms        INTEGER NOT NULL
  model_id          UUID REFERENCES training_runs(id)
  created_at        TIMESTAMPTZ DEFAULT NOW()

```

Indexes are created on events(event_date), events(primary_country, primary_region), event_entities(event_id, entity_type), event_entities(canonical_form) for actor-centric queries, and training_runs(user_id, started_at desc) for the dashboard.

Annex E: Sample Augmentation Templates

A representative subset of the augmentation templates used to generate synthetic training examples. The full lexicon and template catalogue are in backend/scripts/augment_training_data.py.

Verb lexicons

Location-taking action verbs (ACTOR + ACTION + LOCATION pattern): attacked, raided, stormed, invaded, struck, hit, overran, sacked, bombed, shelled, destroyed, burned, torched, razed, demolished, devastated, ravaged, gutted, wrecked, ruined, captured, seized, occupied, surrounded, encircled, besieged, blockaded, conquered, looted, ransacked, pillaged, plundered, breached, sabotaged, vandalised.

Victim-taking action verbs (ACTOR + ACTION + VICTIM pattern): killed, murdered, slaughtered, massacred, executed, shot, butchered, beheaded, decapitated, hanged, lynched, strangled, drowned, poisoned, wounded, injured, maimed, kidnapped, abducted, detained, arrested, apprehended, tortured, brutalised, assaulted, displaced, expelled, evicted.

Clash verbs (ACTOR + CLASH_ACTION + ACTOR pattern): clashed with, skirmished with, exchanged fire with, traded fire with, battled, fought, engaged, confronted, repelled, routed, defeated, overpowered.

Template patterns

```

{actor} {action} {location}, killing {n_killed} {victim}.
{actor} {action} {location}, leaving {n_killed} dead and {n_injured} injured.
{date}, {actor} {action} {location}.
{actor} {action} {location} and looted several buildings.
{n_killed} {victim} were killed when {actor} {action} {location}.
{actor} armed with heavy weapons {action} {location}, resulting in
{n_killed} casualties.
{actor} {action} several villages in {region}, leaving at least
{n_killed} dead.

```

{actor} {action} {location} and abducted {n_killed} {victim}.
{actor} {action} {location} in a dawn raid, killing {n_killed} {victim}.
{actor} seized {location} after a prolonged siege.
{actor} {action} {n_killed} {victim} in {location}.
{date}, {actor} {action} {n_killed} {victim} in {location}.
{actor} {action} at least {n_killed} {victim} in {region}.
{actor} {clash_action} {actor2} in {location}.
{actor} {clash_action} {actor2} in {location}, leaving {n_killed} dead.
Heavy fighting erupted when {actor} {clash_action} {actor2} in {region}.

Date forms

On January 15, 2024, On Monday, Last Tuesday, January 15, Earlier this week, On Wednesday morning, On Friday night.

Annex F: User Acceptance Testing Questionnaire

The questionnaire reproduced here was administered after each user acceptance testing session.

Demographic information

1. Role: ☐ Early-warning analyst ☐ Conflict researcher ☐ Software developer ☐ Other (specify)
2. Years of experience in your role: ____
3. Familiarity with NLP systems: ☐ None ☐ Basic ☐ Intermediate ☐ Advanced

Tasks

For each of the six supplied tasks, please record: - Time to first meaningful interaction (seconds). - Whether you completed the task: ☐ Yes ☐ Yes, with help ☐ No. - Any issues you encountered.

Task list: 1. Paste a supplied article and view the extracted entities. 2. Browse the event store and find events from a specified country. 3. Run an analytics query for events in the past 30 days. 4. Start a new training run with the supplied dataset. 5. Monitor the training-run progress until completion. 6. Review a flagged event and either confirm or edit the extraction.

Likert-scale evaluation

For each statement, indicate your level of agreement on a 1 to 5 scale, where 1 means strongly disagree and 5 means strongly agree.

1. The extracted entities matched what I expected.
2. The 5W1H structuring was clear and easy to interpret.

3. The confidence scores were useful for triage.
4. The knowledge-base enrichment (canonical names, country lookups) added value beyond raw extraction.
5. The training screen made it easy to start and monitor a run.
6. The analytics views answered the kinds of questions I would typically ask.
7. The taxonomy assignment matched my classification of the event.
8. The system response times felt acceptable for my use case.

Open-ended questions

1. What did you find most useful about the system?
2. What did you find most frustrating?
3. What additional features would most improve your workflow?
4. Would you adopt this system in your work, given current limitations? ☐ Yes ☐ With improvements ☐ No. Reason: _____

Signed Declaration Sheet

I, the undersigned, declare that this thesis is my original work and has not been presented for a degree in any other university, and that all source of materials used for the thesis have been duly acknowledged.

Declared by:

Name: _____

Signature: _____

Date: _____

Confirmed by advisor:

Name: _____

Signature: _____

Date: _____